HyperSINet: A Synergetic Interaction Network Combined With Convolution and Transformer for Hyperspectral Image Classification

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***Abstract*— In hyperspectral images (HSIs), both local and nonlocal features play crucial roles in classification tasks. Vision transformers (VITs) can extract nonlocal features through attention mechanisms, while convolutional neural networks (CNNs) excel at handling local components. However, in traditional dual-branch models based on VIT and CNN, there is a lack of interaction during feature processing, leading to potential compatibility issues when merging the two types of features. In this article, we propose HyperSINet, a synergetic interaction network that combines VIT and CNN to establish interaction between the two branches, enabling mutual compensation between local and nonlocal features during the training process and ultimately enhancing the performance of classification tasks. Specifically, we devise a pair of interactors, namely, Conv2Trans and Trans2Conv, which serve as intermediaries between the two branches, enabling the VIT branch to refine its local details, while allowing the CNN branch to process larger receptive field nonlocal features. Typical feature maps are implemented to visualize the function of the interactors. Furthermore, within the VIT branch, a VIT encoder with the local mask is developed to strike a balance between emphasizing nonlocal features and preserving local details, while a lightweight CNN block is designed to process spectral and spatial features in the CNN branch. Extensive experiments conducted on four real-world datasets demonstrate that, under a reasonable count of parameters, HyperSINet surpasses several current state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HIS) classification, interactors, synergetic interaction, vision transformer (VIT).**

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I. Introduction

HYPERSPECTRAL images (HSIs) are a special type of remote sensing imagery that contain several times more spectral information than standard RGB images [1]. Due to the rich information contained in HSIs, wide applications have been seen in fields, such as agriculture, geological exploration, urban planning, and more [2], [3], [4], [5], [6], [7]. The classification of HSIs, being a crucial underpinning of these applications, has garnered widespread attention, aiming to discern the land cover types of all pixels in HSIs [8], [9], [10].

The HSIs classification has been studied for several decades. Due to the limitations of traditional machine learning (ML) methods in the early days, the task was frequently split into two phases, feature extraction and classification. Currently, various methods are utilized for feature extraction, such as principal component analysis (PCA) [11], linear discriminant analysis (LDA) [12], [13], [14], and wavelet transform [15], [16], [17]. For the final classification, k-nearest neighbor (KNN) [18], support vector machine (SVM) [19], and Bayesian estimation [20], [21] have been proposed as effective methods. Despite some successes achieved by ML methods in this domain, they still confront several challenges [22], [23], [24], [25]. ML methods struggle to fully extract abundant information from complex HSIs features. Moreover, defining multiple hyperparameters can pose substantial challenges, requiring numerous manual experiments and specialized expertise [26].

In recent years, the rise of deep learning (DL) methods has significantly improved the performance of HSIs classification tasks. Classic DL methods, such as multilayer perceptron (MLP) [27], recurrent neural networks (RNNs) [28], and convolutional neural networks (CNNs) [29], have been developed to model complex nonlinear relationships in HSIs. Due to its powerful spatial–spectral feature extraction capabilities and outstanding scalability, CNN has become the fundamental model for HSIs classification tasks. Proposed by Roy et al. [30], HybridSN is a hybrid network that extracts discriminative features along the spectral dimension using 3D-CNN and learns spatial features using 2D-CNN. An end-to-end spectral–spatial residual network (SSRN), designed by Zhong et al. [31], takes 3-D cubes of HSIs as input and utilizes

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spectral and spatial residual blocks to extract discriminative features from both dimensions. Swalpa et al. [32] find that previous CNN models usually use fixed-size receptive fields to process the rich spatial–spectral information in HSIs, making it difficult to balance the correlation among the spectral domains. Despite the remarkable achievements of CNN in HSIs classification, there are still inevitable issues. Specifically, convolution operations make it difficult for CNN to extract nonlocal features from HSIs [33]. While it is possible to increase the receptive field of the CNN by deepening the network, to handle nonlocal features, excessive depth can lead to training difficulties and the issue of feature overabstraction, ultimately resulting in model degradation [34].

In 2017, transformer [35] was proposed for machine translation, achieving better performance than traditional models, such as RNN. Vision transformer (VIT) [36], derived from the transformer, adopts a transformer encoder-like structure to process images. By replacing convolution operations with attention mechanisms, VIT significantly enhances its modeling capability for nonlocal features, leading to outstanding performance on visual tasks [37]. In the field of HSIs classification, VIT has given rise to a multitude of exemplary models. Qin and Zhao [38] propose a transformer-based spatial–spectral-associative contrastive learning classification method to address the issue of class imbalance in labeled samples for HSIs classification. SpectralFormer, proposed by Hong et al. [39], is a framework that rethinks HSIs from a sequential perspective with VIT. It is capable of learning spectral sequence relations from adjacent frequency bands of HSIs and employs cross-layer skip connections for adaptive fusion between different layers. Additionally, inputs at both pixel and patch levels can be flexibly accepted by SpectralFormer. To address the issue of inaccurate embedding of labels generated from a single spectrum or spatial dimension, Zou et al. [40] proposed LESSFormer based on VIT, which further enhances the expressive power of tokens. Yang et al. [17] proposed GCN and transformer fusion network (GTFN) based on VIT and graph knowledge, leveraging contextual information of classification pixels while establishing remote dependencies in the spectral domain. While these methods have achieved significant success, VIT-based approaches tend to incorporate numerous unrelated factors when capturing long-range dependencies, thereby diminishing the significance of local features and disregarding spatial prior knowledge.

Contrastingly, spectral–spatial feature tokenization transformer (SSFTT) proposed by Sun et al. [41] first employs convolution layers to extract low-level spectral–spatial features, followed by feature representation and learning using an encoder in VIT. After using CNN for local feature extraction, SSFTT utilizes VIT for expressing nonlocal features. However, such a serial structure cannot balance local and nonlocal features well, resulting in feature degradation and insufficient learning. CTMixer, proposed by Zhang et al. [42], is a deep aggregation framework for HSIs classification, incorporating a dual-branch parallel structure with CNN and VIT to effectively capture both local and nonlocal features. Furthermore, convolution operations are introduced into the multihead attention (MHA) mechanism to further improve classification accuracy. Zhao et al. [43] also approach HSIs feature extraction from both local and nonlocal perspectives and propose a model called CTFSN, which utilizes two fusion methods of addition and stacking to extract features. The backbone of CTFSN also follows a dual-branch parallel structure with CNN and VIT. CTMixer and CTFSN both use a dual-branch parallel structure, which allows for a more comprehensive extraction of local and nonlocal features, thereby enhancing classification performance. Notably, in both of the aforementioned models, the fusion of local and nonlocal features occurs only at the end of the model, with no interaction during the feature processing stages. Conformer introduced by Peng et al. [44], and the mobile-former proposed by Chen et al. [45] have undergone extensive validation regarding the role of interactors. However, it is worth noting that their selected tasks have typically revolved around image classification and object detection. In the domain of HSIs classification, there is currently no precedent for the utilization of interactors to address the shortcomings of dual-branch models. By using two branches separately to process local and nonlocal features, feature loss and changes are inevitable during the processing. As a result, when the two branches are eventually fused, there is a potential risk of encountering latent compatibility issues.

Due to the inherent challenges of solely CNN-based models in modeling nonlocal features, and the difficulty of refining local details in pure VIT-based models, traditional dual-branch models that integrate CNN and VIT lack sufficient interaction in handling local and nonlocal features, resulting in inadequate feature fusion. Therefore, we design a model called HyperSINet, a dual-branch synergistic interaction network based on a combination of the CNN block and VIT encoder. Both the CNN block and VIT encoder are thoughtfully designed. The CNN block captures features from spectral and spatial perspectives, while the VIT Encoder efficiently manages local details using a local mask. Within HyperSINet, we introduce two interactors, Conv2Trans and Trans2Conv, to swiftly convert local and nonlocal features and facilitate mutual integration during processing. In the model, first, the original HSIs inputted into the model undergo shallow feature processing via a two-layer 1×1 convolution block, resulting in the input data for the CNN branch. Second, due to the time complexity of the VIT encoder, inputting all the pixels in HSIs as tokens into the VIT branch demands substantial computational resources. Therefore, we employ superpixels as a substitution for pixels to reduce computational demands. Through establishing a correspondence between pixels and superpixels, an association matrix is obtained for multiplication with the processed HSIs to generate input tokens for the VIT branch. Third, the CNN and VIT branches concurrently process their respective inputs, extracting both local and nonlocal features. During the processing, the local and nonlocal feature conversion is accomplished by the interactors, which take place within each group of CNN block and VIT Encoder. Finally, after processing through the weighted fusion module (FM), each pixel classification is accomplished by a linear classifier consisting of MLP.

The main contributions of this article are summarized as follows.

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1) We design two interactors, Conv2Trans and Trans2Conv, to enable swift mutual transformation between local and nonlocal features, along with soft-selective weighted fusion, thereby enabling the complementary integration of both types of features during feature processing.

2) We develop a CNN block capable of effectively processing local features from both spectral and spatial perspectives. Additionally, we optimize the traditional VIT encoder with the local mask, allowing it to balance nonlocal features while preserving local details.

3) We propose a novel dual-branch synergetic interaction network for HSIs classification, called HyperSINet. HyperSINet achieves mutual embedding of the two branches' features by interaction, enhancing the model's feature processing capability and effectively improving classification performance.

4) We conduct extensive ablation studies and evaluate the performance of HyperSINet on four representative real HSIs datasets, i.e., Indian Pines, Pavia University (PaviaU), Salinas, and Houston 2013. The experimental results demonstrate that HyperSINet exhibits significant advantages over state-of-the-art models in HSIs classification.

The organization of the remainder of this article is as follows. Section II presents the methodology of HyperSINet. Section III introduces the datasets, showcasing the classification performance of HyperSINet compared with other classifiers. Section IV provides a comprehensive analysis of the hyperparameter settings through experiments. Additionally, ablation experiments are performed to further investigate the roles of each module in HyperSINet. Finally, Section V summarizes this work.

II. Proposed Method

The overall architecture of HyperSINet proposed in this article is shown in Fig. 1. At the core of HyperSINet lies a dual-branch parallel structure network comprised of CNN block and VIT encoder. Two operations are carried out on the original HSIs input within HyperSINet. On the one hand, dimensional adjustments are performed by a two-layer 1×1 convolution block, achieving shallow-level feature processing, and the resulting output is utilized as the input to the CNN branch. On the other hand, pixel-to-superpixel transformation is performed, resulting in the derivation of an association matrix representing the pixel-superpixel correspondence. The input tokens for the VIT branch are obtained through the multiplication of the association matrix with the processed HSIs data. In contrast to the conventional VIT model, positional embedding and learnable token concating operations are omitted. During the dual-branch network processing, a pair of interactors is proposed to facilitate rapid mutual conversion between the local features from the CNN branch and the nonlocal features from the VIT branch. Ultimately, the outputs from the two branches undergo processing through a weighted FM and achieve final classification via a linear classifier.

A. Initial HSIs Preprocessing

Suppose $\hat{X} \in \mathbb{R}^{C\times H\times W}$ represents the initial HSIs data, where C is the number of bands, H is the height, and W is the width. Each pixel in $\hat{X}$ contains C spectral measures. Resulting of mixed land cover categories, a great deal of spectral redundancy appears in HSIs. Therefore, a two-layer 1×1 convolution block is utilized for the removal of redundant spectral bands and dimension adjustment. In each layer, a BatchNormalization2D, followed by a 1×1 convolution operation and a LeakyReLU activation function, are included. The HSIs after processing are denoted as $X \in \mathbb{R}^{B\times H\times W}$, where B represents the number of bands after removing redundancy. In this work, B is defined as 64. The process of obtaining X is outlined as follows:

$$X = \Phi^2\left(\hat{X}\right) \tag{1}$$

where Φ represents a 1×1 convolution block. X serves as the input data for the CNN branch.

Inputting all pixels from HSIs as tokens into the VIT encoder places an extremely high demand on computational resources. Hence, simple linear iterative clustering (SLIC) [46] is employed to obtain superpixels, reducing the total number of tokens. Based on the correspondence between pixels and superpixels, an association matrix $Q \in \mathbb{R}^{HW\times Z}$ is obtained, where Z denotes the number of superpixels. The process is depicted in Fig. 2. Q is a binary matrix, it can be represented by the following formula, where $Q_{i,j}$ denotes the value at position (i, j) in Q, and $\widetilde{X}$ is the pixel feature set obtained by flattening X along the spatial dimensions. $\widetilde{X}_i$ corresponds to the ith pixel, and S_j refers to the jth superpixel

$$Q_{i,j} = \begin{cases} 1, & \widetilde{X}_i \in S_j \\ 0, & \widetilde{X}_i \notin S_j \end{cases} \quad \widetilde{X} = \text{Flatten}(X). \tag{2}$$

As illustrated in Fig. 1, Q is obtained from the $\hat{X}$ through the process of SLIC. However, the superpixel features are derived from the multiplication of X and Q. Employing the derivation of Q from X, this process is repeated in each epoch, incurring significant time overhead. Therefore, after careful consideration, the decision is made to preprocess the SLIC before model training. This entails the acquisition of Q, which is subsequently used to perform the multiplication with X, thereby generating superpixel features as input tokens for the VIT branch. The process of obtaining tokens $X_V \in \mathbb{R}^{Z\times B}$ is illustrated as follows:

$$X_V = \hat{Q}^T \widetilde{X} \tag{3}$$

where $\hat{Q}$ represents the row-normalized result of Q. As illustrated in Fig. 2(e), to accomplish subsequent pixel-level classification tasks, the restoration of X_V to pixel-level features is still facilitated through the utilization of Q, the process is illustrated as follows:

$$\widetilde{X} = QX_V. \tag{4}$$

B. Local Mask-Optimized VIT Encoder

Traditional VIT encoder uses a global attention mechanism to calculate intertoken attention scores, but the global modeling approach often overlooks local spatial details, limiting the ability to refine local features. Inspired by LESSFormer proposed by Zou et al. [40], we introduce a local mask to enhance

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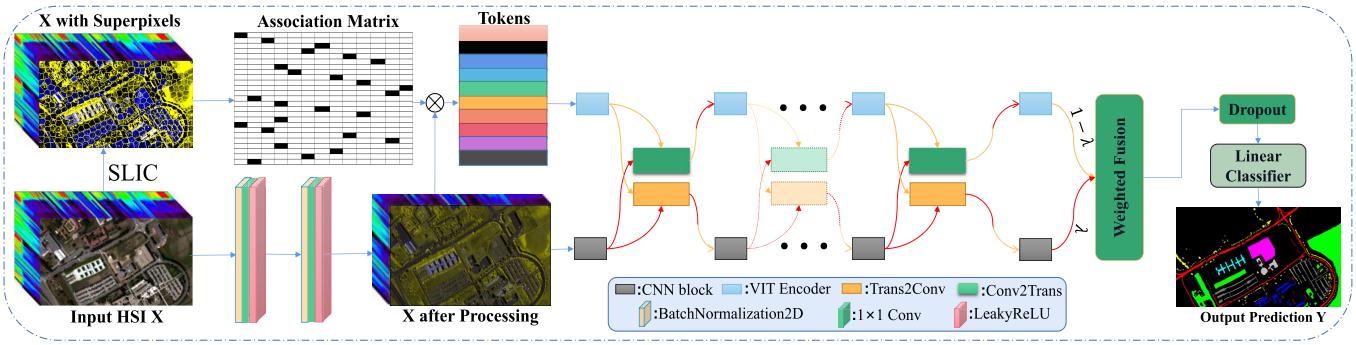


Fig. 1. Overall structure of the proposed HyperSINet for HSIs classification.

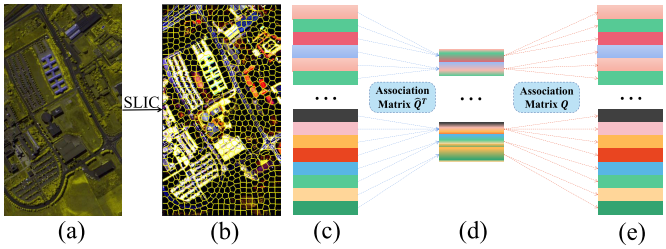


Fig. 2. Illustration of the SLIC and Associated Matrix effects. (a) PaviaU with pixels. (b) PaviaU with superpixels. (c) Pixel features after flattening. (d) Superpixel features. (e) Restored pixel features.

the local feature refinement capabilities of the conventional VIT encoder. The localized mask-optimized VIT encoder (LMOVE) is illustrated in Fig. 3. LMOVE is composed of an MHA layer, two layer normalization (LN) layers, and an MLP block. The MHA plays a pivotal role in the LMOVE, with the local mask-enhanced attention (LMEA) being the crucial component within it. The overall structure of MHA and LMEA is illustrated in Fig. 3(b) and (c), respectively. In LMEA, three trainable parameter matrices, namely, $\hat{W}_q \in \mathbb{R}^{B\times B}$, $\hat{W}_k \in \mathbb{R}^{B\times B}$, and $\hat{W}_v \in \mathbb{R}^{B\times B}$, are designed to capture diverse correlations between the features. Three vectors query $q \in \mathbb{R}^{Z\times B}$, keys $k \in \mathbb{R}^{Z\times B}$, and values $v \in \mathbb{R}^{Z\times B}$ are obtained through the transformation of the three-parameter matrices. The attention scores between vectors q and k are calculated through the inner product. However, the attention scores primarily convey a global perspective, neglecting the significance of local details. Thus, we propose a local mask [shown in Fig. 4(b)] based on superpixel adjacency to enhance local detail descriptions by reducing attention between nonadjacent superpixels. Unlike the SNN mask in LESSFormer, the proposed local mask adopts a weighted adjacency matrix format normalized by rows to measure the attention level between superpixels using weights. The outcome of applying the local mask to the attention score matrix using Hadamard product is activated with the sigmoid activation, transforming positions with zero value in the local mask into infinitesimal positive values. Consequently, attention between nonadjacent superpixels is effectively suppressed, while attention to adjacent superpixels is retained and amplified. The process can be represented as follows:

$$A_L = \text{Sigmoid}\left(\widetilde{M}_L \odot qk^T\right)v \tag{5}$$

where $A_L \in \mathbb{R}^{Z\times B}$ denotes the output processed after the sigmoid activation, and $\odot$ represents the dot-product operation. $\widetilde{M}_L$ is the local mask, which is derived from the weighted adjacency matrix M_L that is normalized by rows. The specific representation of M_L is as follows:

$$M_{L_{i,j}} = \begin{cases} e^{-\beta\|S_i - S_j\|_2^2}, & i = j \text{ or } S_i \text{ is adjacent to } S_j \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where β denotes the steepness hyperparameter affecting the weights.

A_L transforms three additional trainable matrices, denoted as $M_{q'} \in \mathbb{R}^{B\times B}$, $M_{k'} \in \mathbb{R}^{B\times B}$, and $M_{v'} \in \mathbb{R}^{B\times B}$, resulting in three new vectors, namely, q', k', and v'. Attention is computed using q' and k', followed by elementwise multiplication with the nonlocal mask [shown in Fig. 4(c)]. Subsequently, global attention weights are obtained through Softmax activation. The process can be elucidated as follows:

$$A_N = \text{Softmax}\left(\widetilde{M}_N \odot \frac{q'k'^T}{\sqrt{d}}\right)v' \tag{7}$$

where $A_N \in \mathbb{R}^{Z\times B}$ is the output of LMEA, $\widetilde{M}_N$ is the nonlocal mask, d denotes the dimension of k', and $\odot$ represents the dot-product operation. In the MHA, multiple sets of LMEA are used to compute various attention values. These results are concatenated and processed by a linear transformation matrix to adjust the output dimensions to match the input data.

C. CNN Block

The CNN block, illustrated in Fig. 5, processes local features from both spectral and spatial perspectives. It consists of four parts, which are the transition module (TM), spectral feature processing module (SpeM), spatial feature processing module (SpaM), and FM.

1) Transition Module: The TM is designed to make internal data adjustments using very few parameters while keeping the dimensions unchanged. Pointwise convolution (PwConv) and depthwise convolution (DwConv) are introduced to maintain a reduced parameter count. Within a CNN block, two TMs are co-located, and their differentiation is contingent upon the disparate convolution kernel sizes in DwConv, specifically, the

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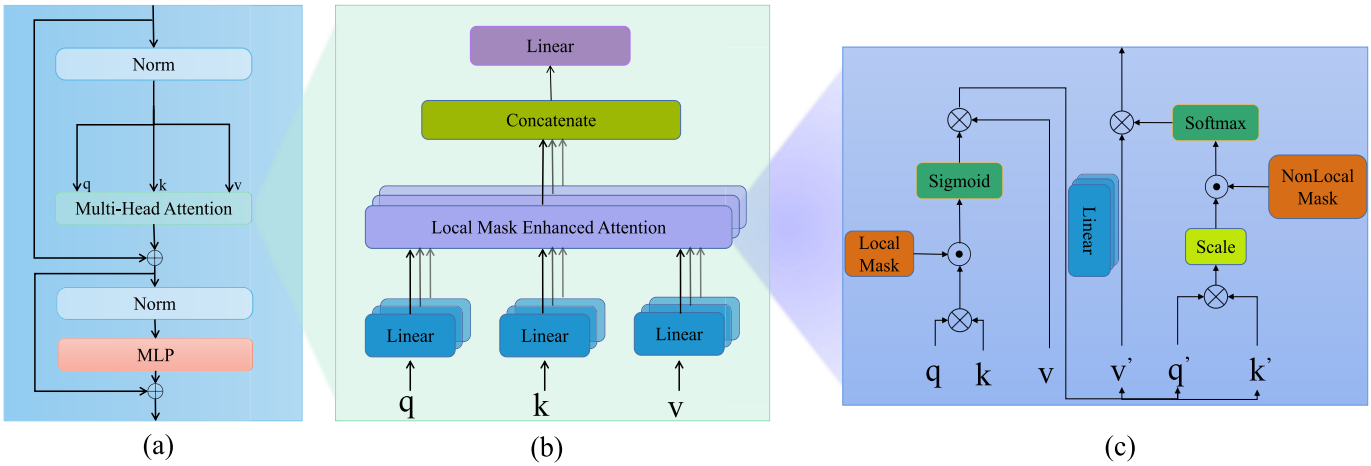


Fig. 3. Overall structure of the local mask-optimized VIT encoder. (a) Local mask-optimized VIT encoder. (b) MHA. (c) LMEA.

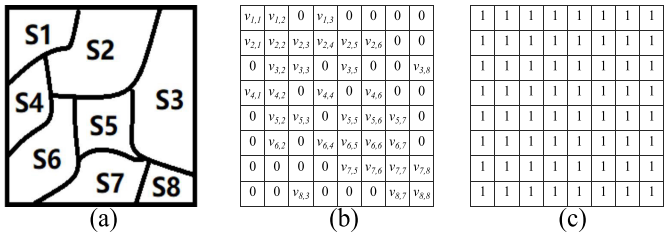


Fig. 4. Illustration of two types of masks. (a) Image with Superpixels. (b) Local mask. (c) Nonlocal mask.

first TM employs a 3×3 convolution kernel in DwConv, whereas the second utilizes a 5×5 kernel. The procedure for obtaining the output $O_{\mathrm{TM}} \in \mathbb{R}^{B\times H\times W}$ of TM is as follows:

$$O_{\mathrm{TM}} = \delta(f_{Dw}(\delta(f_{Pw}(\phi(X))))) \tag{8}$$

where δ denotes LeakyReLU activation function, $f_{Dw}(\cdot)$ is the DwConv, and ϕ is the BatchNormalization2D function.

2) Spectral Feature Processing Module: SpeM is designed for spectral feature processing. Initially, two 1×1 convolutions are applied to achieve dimension reduction, resulting in $D_1 \in \mathbb{R}^{1\times H\times W}$ and $D_2 \in \mathbb{R}^{(B/2)\times H\times W}$. D_1 and D_2 are subjected to flattening along their spatial dimensions, and D_2 is transposed, leading to the adjusted dimensions of $[1, HW]$ and $[HW, (B/2)]$. After Softmax processing, D_1 is multiplied by D_2 and expand the result by one dimension to obtain $D_3 \in \mathbb{R}^{(B/2)\times 1\times 1}$. Utilizing a 1×1 convolution for dimension adjustment, D_3 is transformed to obtain $D_4 \in \mathbb{R}^{B\times 1\times 1}$. Multiplying D_4 by the input O_{TM} of SpeM to modulate the spectral information, yields the output $O_{\mathrm{Spe}} \in \mathbb{R}^{B\times H\times W}$ of SpeM. The acquisition process of O_{Spe} is depicted as follows:

$$O_{\mathrm{Spe}} = f^{B}_{1\times 1}\left(\xi\left(\mathrm{Softmax}\left(\Delta\left(f^{1}_{1\times 1}(X)\right)\right)\Delta\left(f^{\frac{B}{2}}_{1\times 1}(X)\right)^{T}\right)\right)X \tag{9}$$

where $\Delta(\cdot)$ denotes the operation of flattening, $\xi(\cdot)$ represents the operation of expanding dimension, and $f^{B}_{1\times 1}(\cdot)$ signifies a convolution with a 1×1 kernel, yielding an output dimension of B.

3) Spatial Feature Processing Module: SpaM is designed for spatial feature processing. Like SpeM, 1×1 convolution is used to adjust the spectral dimension to obtain $D_5 \in \mathbb{R}^{(B/2)\times H\times W}$ and $D_6 \in \mathbb{R}^{(B/2)\times H\times W}$. Global average pooling (AvgPool) is applied to D_5 and transposed to obtain $D_7 \in \mathbb{R}^{1\times(B/2)}$. D_6 is flattened along the spatial dimensions, yielding $D_8 \in \mathbb{R}^{(B/2)\times HW}$. D_8 is multiplied by Softmax processed D_7, resulting in $D_9 \in \mathbb{R}^{1\times HW}$. After undergoing reshaping and dimension expansion, the dimension of D_9 becomes $[1, H, W]$, preserving valuable spatial information, and then multiplied by X to produce the output $O_{\mathrm{Spa}} \in \mathbb{R}^{B\times H\times W}$ of SpaM. The process can be represented as follows:

$$O_{\mathrm{Spa}} = \xi\left(\widetilde{\Delta}\left(\mathrm{Softmax}\left(\sigma\left(f^{\frac{B}{2}}_{1\times 1}(X)\right)^{T}\right)\Delta\left(f^{\frac{B}{2}}_{1\times 1}(X)\right)\right)\right)X \tag{10}$$

where $\widetilde{\Delta}(\cdot)$ represents the operation of reshaping, and $\sigma(.)$ denotes the AvgPool.

4) Fusion Module: O_{Spe} and O_{Spa} have rich spectral and spatial information. Basic addition cannot fully combine them, so FM is designed for adaptive fusion. In FM, $D_{10} \in \mathbb{R}^{2B\times H\times W}$ is formed through the concatenation of O_{Spe} and O_{Spa} in spectral dimension. Each channel of D_{10} undergoes global max-pooling (MaxPool) and AvgPool, yielding two distinct feature distributions referred to as $D_{11} \in \mathbb{R}^{2B\times 1\times 1}$ and $D_{12} \in \mathbb{R}^{2B\times 1\times 1}$. D_{11} and D_{12} are spatially concatenated, followed by their fusion through a 2×1 convolutional, and then processed through Softmax to obtain $D_{13} \in \mathbb{R}^{2B\times 1\times 1}$. D_{13} contains weight coefficients for the channels of D_{10}. Multiplying them with D_{10} yields weighted features. Adding the weighted features to D_{10} through skip connection produces the $D_{14} \in \mathbb{R}^{2B\times H\times W}$. In the end, a 1×1 convolution is applied to D_{14} to capture interchannel relationships and adjust the spectral dimensions, yielding the output $O_{\mathrm{FM}} \in \mathbb{R}^{B\times H\times W}$ of FM. This process can be represented as follows:

$$\begin{aligned} O_{\mathrm{Cat}} &= \left[O_{\mathrm{Spe}}, O_{\mathrm{Spa}}\right] \\ D_{13} &= \mathrm{Softmax}\left(f^{2B}_{2\times 1}([\sigma(O_{\mathrm{Cat}}), \psi(O_{\mathrm{Cat}})])\right) \\ O_{\mathrm{FM}} &= f^{B}_{1\times 1}(D_{13}O_{\mathrm{Cat}} + O_{\mathrm{Cat}}) \end{aligned} \tag{11}$$

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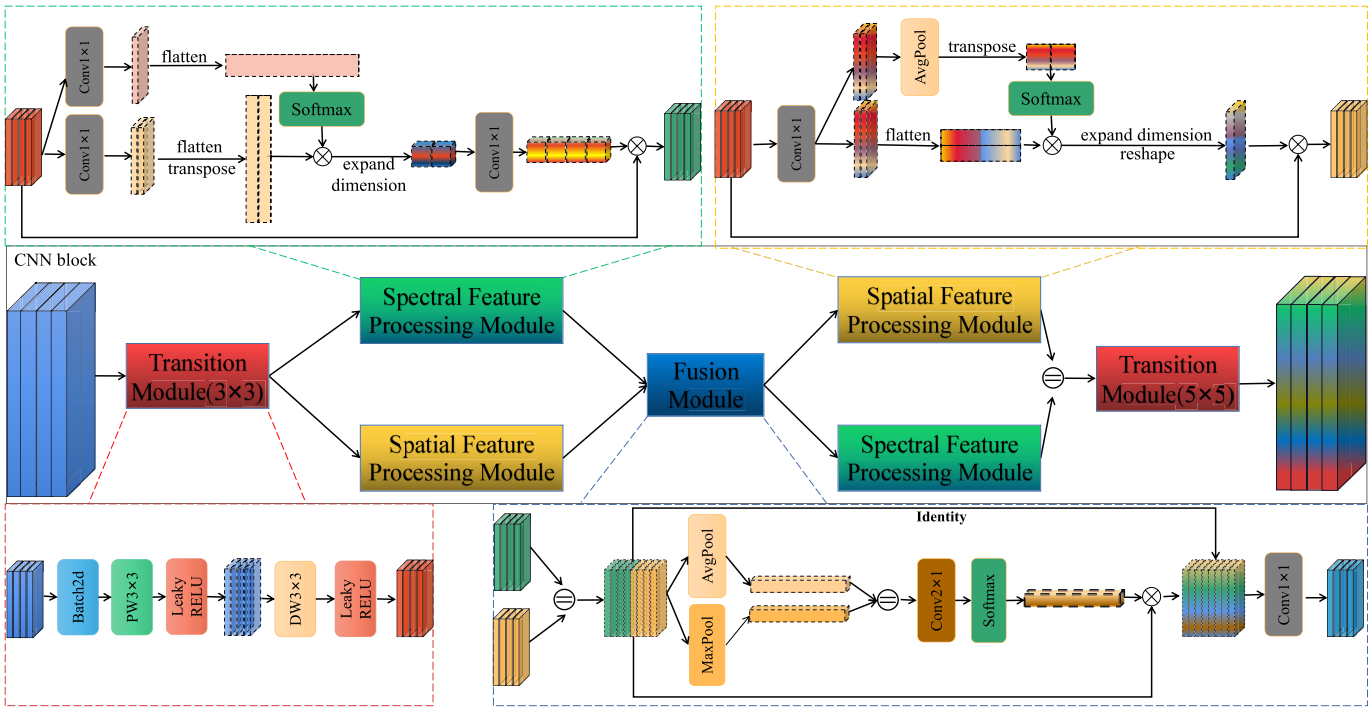


Fig. 5. Overall structure of the CNN block.

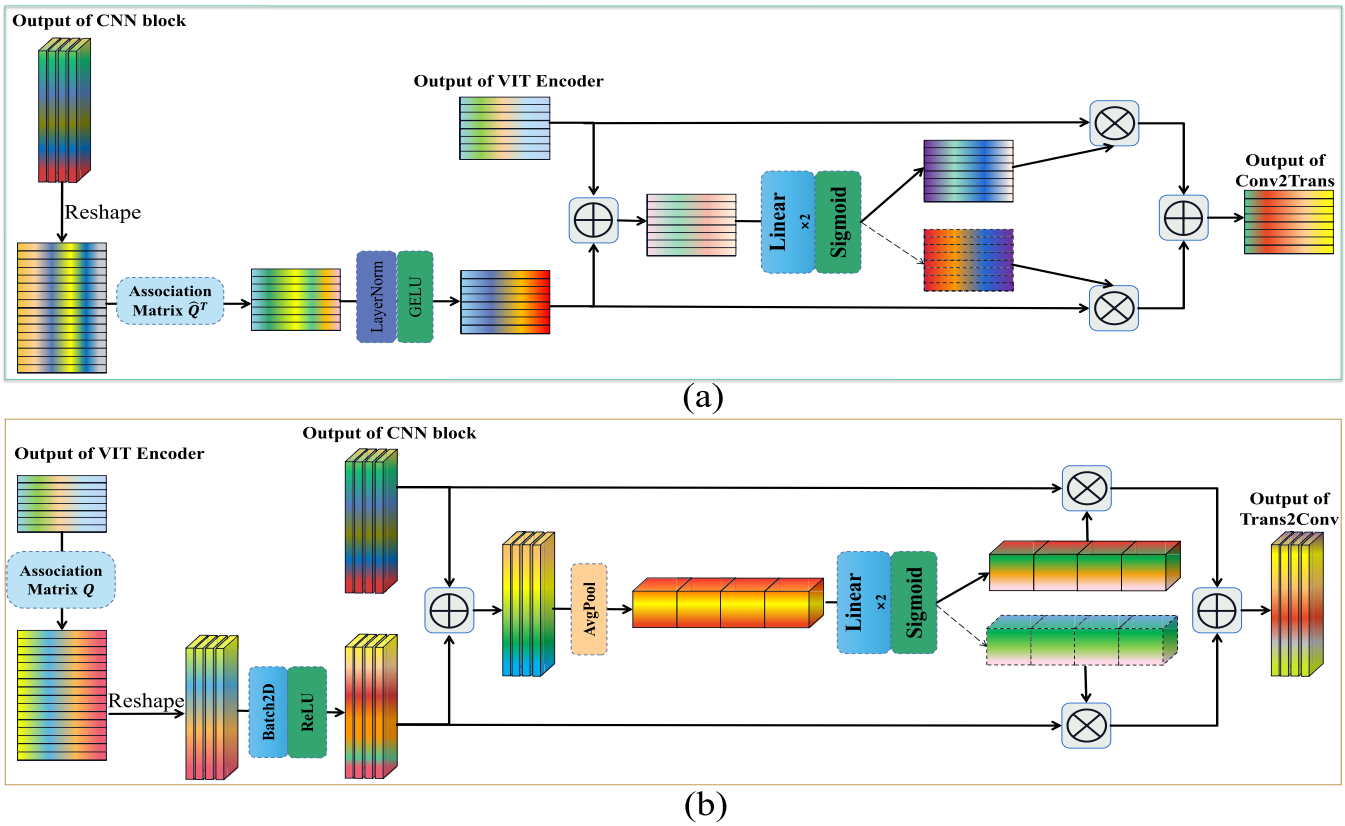


Fig. 6. Overall structure of the two Interactors. (a) Conv2Trans. (b) Trans2Conv.

where $\psi(\cdot)$ denotes the MaxPool operation and $[,]$ is the operation of concatenation. The output O_{FM} undergoes separate processing through SpaM and SpeM, and their respective output results are concatenated along the spectral dimension. The transition is then completed in the second TM, and the dimension of output is adjusted to $[B, H, W]$.

D. Conv2Trans and Trans2Conv

In conventional dual-branch networks for HSIs classification, a deficiency in interactivity is observed between the two branches. They only interact at the final feature fusion stage, leading to noticeable differences in the output features due to their distinct focuses [44], [47], [48]. Hence, we design

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two interactors, namely, Conv2Trans and Trans2Conv, allowing the VIT branch to learn detailed local features from the CNN branch, and enabling the CNN branch to capture broader nonlocal features from the VIT branch. The overall structure is depicted in Fig. 6. In each interactor, the output features of both branches are utilized as their respective input, enabling the transformation and fusion of two types of features within the interactor. To ensure the utmost feature integrity, no trainable parameters are introduced during the feature transformation stage, and the stage is achieved solely through the association matrix. During feature fusion, a more flexible and reliable fusion framework based on sigmoid is introduced into both interactors, mitigating issues of feature incompatibility associated with direct addition or elementwise multiplication. Specifically, the feature processing procedures within the two interactors are as follows.

1) Conv2Trans: The architecture of Conv2Trans is illustrated in Fig. 6(a). The input, $I_C \in \mathbb{R}^{B\times H\times W}$, from the CNN branch undergoes a reshape operation, flattening into a 2-D vector with dimensions $[HW, B]$. To obtain nonlocal feature embeddings, $\hat{Q}^T \in \mathbb{R}^{Z\times HW}$ is subjected to matrix multiplication with the 2-D vector, resulting in a dimensional transformation to $[Z, B]$. LN and GELU are employed to normalize the internal data distribution of the vector, resulting in the final feature embeddings, denoted as $I_{c2t} \in \mathbb{R}^{Z\times B}$. During the feature fusion stage, the elementwise summation of I_{c2t} and I_V is chosen as the initial integration. Two linear layers are introduced to adjust the internal feature distribution of the initial integration, thereby enhancing the robustness of the fusion process. The output of the linear layers is mapped to real numbers between 0 and 1 through the sigmoid function, serving as fusion weights $SD(\text{Linear}^2(I_{c2t}+I_V))$. Meanwhile, the weight associated with the dashed portion in Fig. 6(a) is expressed as $1 - SD(\text{Linear}^2(I_{c2t} + I_V))$, which enable the network to conduct a soft selection or weighted averaging between I_{c2t} and I_V. The procedure for obtaining the output O_{c2t} of Conv2Trans is illustrated as follows:

$$\begin{aligned} I_{c2t} &= \Psi\left(\varphi\left(\hat{Q}^T\widetilde{\Delta}(I_C)\right)\right) \\ \text{Wei}_{c2t} &= SD\left(\text{Linear}^2(I_{c2t}+I_V)\right) \\ O_{c2t} &= I_V\text{Wei}_{c2t} + I_{c2t}(1-\text{Wei}_{c2t}) \end{aligned} \tag{12}$$

where $\Psi(\cdot)$ and $\varphi(\cdot)$, respectively, represent GELU and LN. SD denotes the sigmoid function.

2) Trans2Conv: The architecture of Trans2Conv is illustrated in Fig. 6(b). In order to obtain local feature embeddings, the input $I_V \in \mathbb{R}^{Z\times B}$ undergoes matrix multiplication with $Q \in \mathbb{R}^{HW\times Z}$, resulting in a 2-D vector of dimensionality $[HW, B]$. After the 2-D vector undergoes a reshape operation, it attains the same dimensionality as $I_C \in \mathbb{R}^{B\times H\times W}$. Subsequently, it undergoes batch normalization (BN) and rectified linear unit (ReLU) activation to obtain the final local feature embeddings I_{t2c}. During the feature fusion stage, the elementwise summation of I_{t2c} and I_C is chosen as the initial integration. AvgPool is applied to the integration along the spectral dimension, resulting in a 1-D spectral vector that consolidates spatial features across the entire spectrum. Subsequent operations align with those in Conv2Trans. The procedure for obtaining the output O_{t2c} of Trans2Conv is illustrated as follows:

$$\begin{aligned} I_{t2c} &= \psi\left(\widetilde{\varphi}\left(\widetilde{\Delta}(QI_V)\right)\right) \\ \text{Wei}_{t2c} &= SD\left(\text{Linear}^2(I_{t2c}+I_C)\right) \\ O_{t2c} &= I_C\text{Wei}_{t2c} + I_{t2c}(1-\text{Wei}_{t2c}) \end{aligned} \tag{13}$$

where $\psi(\cdot)$ and $\widetilde{\varphi}(\cdot)$, respectively, represent ReLU and BN.

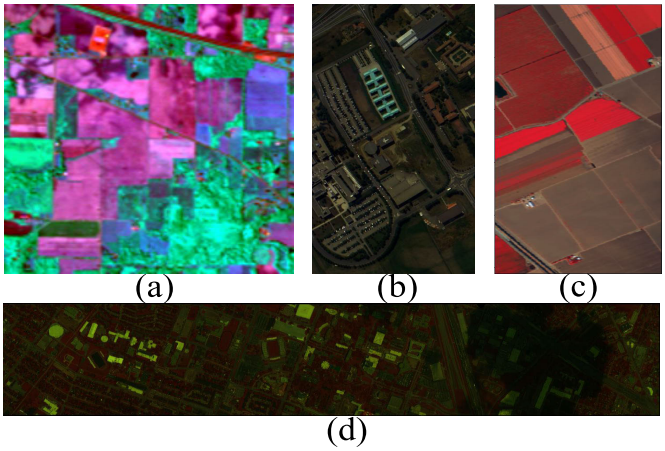


Fig. 7. False-color images of four datasets. (a) Indian Pines. (b) PaviaU. (c) Salinas. (d) Houston.

E. Weighted Fusion and Classification

A weighted FM (WF) is designed to integrate the output features from two branches, with the fusion rate designated as λ. Within the WF, superpixel features from the VIT branch are transformed into pixel features using Q initially. Subsequently, weighted fusion is executed with the CNN branch. The overall process is delineated as follows:

$$O_{\text{All}} = Q(1-\lambda)O_{\text{VIT}} + \lambda O_{\text{CNN}}. \tag{14}$$

A one-layer linear classifier is used to adjust the dimensions of O_{All} for the final classification in HyperSINet. Cross-entropy function serves as the loss function for network training and parameter updates.

III. Experiments

In this section, we first introduce four well-known datasets of HSIs and then compare the experimental performance of HyperSINet and ten other methods on four datasets.

A. Data Description

In this article, four datasets are employed, i.e., Indian Pines, PaviaU, Salinas, and Houston 2013 (Houston), with their false-color images illustrated in Fig. 7. The details of the four datasets are as follows.

1) Indian Pines Dataset: The Indian Pines dataset obtained in 1992 [49]. The original Indian Pines had 220 consecutive bands, but after removing 20 noisy bands, the dataset used in this study was obtained. It consists of 16 categories and has a spatial dimension of $145 \times 145 \times 200$, encompassing a total of 10 249 valid pixels.

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TABLE I
SPECIFIC DETAILS OF THE SPLITS ACROSS FOUR DATASETS

No.	Indian Pines			PaviaU			Salinas			Houston		
	Training.	Val.	Testing.	Training.	Val.	Testing.	Training.	Val.	Testing.	Training.	Val.	Testing.
1	2	2	42	33	33	6565	10	10	1989	63	63	1125
2	71	71	1286	93	93	18463	19	19	3688	63	63	1128
3	42	42	746	10	10	2079	10	10	1956	35	35	627
4	12	12	213	15	15	3034	7	7	1380	62	62	1120
5	24	24	435	7	7	1331	13	13	2652	62	62	1118
6	37	37	656	25	25	4979	20	20	3919	16	16	293
7	1	1	26	7	7	1316	18	18	3543	63	63	1142
8	24	24	430	18	18	3646	56	56	11159	62	62	1120
9	1	1	18	5	5	937	31	31	6141	63	63	1126
10	49	49	874				16	16	3246	61	61	1105
11	123	123	2209				5	5	1058	62	62	1111
12	29	29	535				10	10	1907	62	62	1109
13	10	10	185				5	5	906	23	23	423
14	63	63	1139				5	5	1060	21	21	386
15	19	19	348				36	36	7196	33	33	594
16	5	5	83				9	9	1789			
Total	512	512	9225	213	213	42350	270	270	53589	751	751	13527

2) PaviaU Dataset: The PaviaU dataset is a subset of the data collected from the city of Pavia, Italy, in 2003 [50]. PaviaU initially had 115 spectral bands, but 12 were removed due to noise, resulting in a final set of 103 bands. The spatial dimensions are 610 × 340 pixels, covering nine land cover categories with 42 776 meaningful pixels.

3) Salinas Dataset: The Salinas dataset comprises an image with dimensions of 512 × 217, representing the actual terrain of the Salinas Valley, CA, USA [51]. It contains 204 spectral bands and 54 129 ground object pixels for classification, annotated with 16 different land cover types.

4) Houston Dataset: The Houston dataset was obtained at the University of Houston in 2013 and was initially used for the IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest [52]. The dataset has a size of 349 × 1905 × 144, comprising a total of 15 029 valid pixels.

All four datasets are split into training, validation, and testing sets. Indian Pines and Houston, with a limited number of valid labels, use a 5:5:90 splitting ratio. Meanwhile, the other two datasets with more labels employ a 5:5:990 splitting ratio. The division of the four datasets is presented in Table I.

B. Experimental Setup

1) Evaluation Metrics: To evaluate the performance of the proposed HyperSINet and other state-of-the-art models, four evaluation metrics are employed, i.e., overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and per-class classification accuracy. These metrics are positively correlated with model performance.

2) Comparison With State-of-the-art Models: We compare HyperSINet with state-of-the-art models from different domains to thoroughly evaluate its performance. Among the CNN-based models, we choose multiscale 3D deep CNN (M3D-DCNN) [53], attention-based bidirectional long short-term memory network (AB-LSTM) [54], HybridSN [30], SSRN [31], and A^{2}S^{2}KResNet [55] for comparison. In the context of graph neural networks (GNNs) [56], mini graph convolution network (miniGCN) [10] is chosen. Additionally, we select four VIT-based models, which include the original VIT [36], SpectralFormer [39], SSFTT [41], and group-aware hierarchical transformer (GAHT) [57]. All ten comparative models maintain their original paper configurations, and our reported results are the averages of eight independent runs.

TABLE II
HYPERPARAMETERS USED IN THE EXPERIMENTS

	Hyperparameter	Value
SLIC	Superpixel Scale	500
	Compactness	1
	Max_iter	20
	Sigma	1
	Min_size_factor	0.1
	Max_size_factor	2
Training Parameters	Learning Rate	0.001
	Training Epochs	300
	Optimizer	Adam
	Loss Function	Cross Entropy
	Early Stopping	False
	Weight_decay	0.001
HyperSINet	Spectral Dimension of X	64
	Number of heads	8
	Depth of Network	2
	Fusion Rate	0.05
	Hidden Dim of MLP	64
	Linear1 in interactors	[in_ch,in_ch//2]
	Linear2 in interactors	[in_ch//2,in_ch]

3) Configuration of Hyperparameters: The algorithm proposed is implemented using Python 3.8.0 and PyTorch 1.7.0. All experiments are conducted in an environment comprising an i7-8700K CPU and an NVIDIA A100 Tensor Core GPU. All training, validation, and test sets for all experiments are uniform and prepartitioned based on predetermined proportions, with the allocation of samples into the three sets being randomly determined. The hyperparameters used in the experiments are shown in Table II, where in_ch represents the input dimension.

4) Computational Complexity Analysis: HyperSINet consists of two branches and several modules, with computational

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TABLE III

COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE INDIAN PINES DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	M3D-DCNN	AB-LSTM	A^2S^2KResNet	SSRN	HybridSN	miniGCN	VIT	SpectralFormer	GAHT	SSFTT	HyperSINet
class 1	59.02 ± 21.89	17.30 ± 15.81	**93.37 ± 0.69**	55.00 ± 0.61	68.42 ± 0.53	93.18 ± 2.86	34.96 ± 23.18	22.44 ± 9.18	41.46 ± 14.22	65.37 ± 18.77	92.10 ± 0.83
class 2	77.77 ± 3.89	84.93 ± 2.93	94.63 ± 0.88	95.20 ± 1.00	88.79 ± 0.18	83.76 ± 2.77	82.83 ± 3.07	66.83 ± 3.54	93.20 ± 3.06	95.49 ± 0.37	**96.63 ± 0.47**
class 3	72.56 ± 6.08	65.21 ± 12.43	93.84 ± 0.11	92.85 ± 0.14	88.06 ± 0.42	70.07 ± 1.54	92.73 ± 0.76	56.09 ± 9.76	95.05 ± 3.42	**99.71 ± 0.15**	97.67 ± 1.51
class 4	62.35 ± 8.45	43.47 ± 20.53	96.05 ± 0.91	94.22 ± 0.81	70.87 ± 1.68	79.22 ± 1.93	99.53 ± 0.47	49.86 ± 9.70	89.86 ± 2.23	**99.62 ± 0.84**	98.03 ± 2.81
class 5	88.87 ± 2.45	85.39 ± 3.75	**98.47 ± 0.02**	97.88 ± 0.88	89.16 ± 0.70	84.14 ± 0.99	94.94 ± 2.22	72.78 ± 5.45	93.06 ± 4.10	96.92 ± 0.62	97.92 ± 1.09
class 6	99.21 ± 0.18	95.41 ± 2.46	98.57 ± 0.34	97.81 ± 0.06	97.59 ± 0.77	97.62 ± 0.31	97.46 ± 1.07	87.91 ± 4.30	**99.63 ± 8.36**	99.30 ± 0.25	99.02 ± 0.79
class 7	74.40 ± 28.58	11.82 ± 19.41	79.53 ± 0.10	68.33 ± 0.87	51.52 ± 1.01	92.31 ± 1.07	81.33 ± 22.74	33.60 ± 21.85	72.80 ± 27.53	**100.00 ± 0.00**	94.17 ± 4.73
class 8	98.09 ± 0.92	96.96 ± 1.67	99.30 ± 0.60	97.57 ± 0.21	96.67 ± 1.40	**100.00 ± 0.00**	97.83 ± 1.94	93.91 ± 4.49	99.77 ± 0.29	98.84 ± 0.79	99.49 ± 0.59
class 9	46.67 ± 9.69	00.00 ± 0.00	63.93 ± 2.43	70.06 ± 0.47	**90.00 ± 0.03**	83.33 ± 2.51	59.26 ± 8.49	23.33 ± 11.86	32.22 ± 10.77	76.67 ± 2.48	73.33 ± 4.19
class 10	72.87 ± 6.47	75.04 ± 4.03	93.74 ± 0.04	94.04 ± 1.83	92.70 ± 0.73	81.83 ± 2.47	91.70 ± 0.86	70.99 ± 3.48	91.11 ± 4.20	95.13 ± 2.48	**96.23 ± 1.34**
class 11	85.73 ± 2.55	82.80 ± 3.83	97.37 ± 0.25	96.64 ± 1.27	94.62 ± 0.60	90.52 ± 2.78	95.72 ± 1.00	80.63 ± 3.25	95.55 ± 1.24	**99.49 ± 0.23**	98.94 ± 0.31
class 12	73.56 ± 9.70	58.15 ± 11.76	**97.07 ± 0.16**	96.09 ± 0.52	81.94 ± 0.57	61.96 ± 12.09	81.71 ± 1.70	39.10 ± 6.20	93.90 ± 4.62	90.82 ± 0.83	96.06 ± 2.78
class 13	99.24 ± 0.81	74.51 ± 37.26	98.11 ± 0.31	96.86 ± 0.73	**100.00 ± 0.00**	95.48 ± 0.54	99.28 ± 0.83	98.16 ± 1.11	99.78 ± 0.43	99.35 ± 0.70	99.67 ± 0.73
class 14	92.55 ± 3.63	97.47 ± 0.44	97.55 ± 0.75	97.15 ± 0.11	93.70 ± 0.52	96.13 ± 0.77	97.83 ± 0.98	88.87 ± 5.19	98.03 ± 1.53	98.89 ± 0.51	**99.72 ± 0.29**
class 15	82.48 ± 5.91	60.45 ± 18.60	95.96 ± 0.16	93.74 ± 0.07	86.97 ± 0.18	**100.00 ± 0.00**	89.43 ± 5.41	60.46 ± 6.45	88.76 ± 4.99	97.64 ± 2.17	98.73 ± 1.69
class 16	82.62 ± 16.05	87.03 ± 9.99	91.81 ± 0.16	92.63 ± 0.26	65.00 ± 0.03	**98.90 ± 1.64**	75.40 ± 7.65	97.62 ± 1.30	93.81 ± 2.95	98.33 ± 1.81	96.63 ± 1.96
OA(%)	83.51 ± 2.23	80.47 ± 3.59	96.22 ± 0.02	95.79 ± 0.17	91.07 ± 0.09	87.06 ± 1.01	92.29 ± 0.71	73.72 ± 2.54	94.60 ± 0.66	97.52 ± 0.26	**98.04 ± 0.22**
AA(%)	79.25 ± 3.32	64.75 ± 6.35	93.08 ± 0.98	89.75 ± 1.21	84.75 ± 1.75	88.03 ± 0.89	85.75 ± 1.57	65.16 ± 2.56	86.12 ± 0.89	94.47 ± 1.18	**95.90 ± 0.28**
Kappa × 100	81.11 ± 2.60	77.77 ± 4.08	95.70 ± 0.11	95.20 ± 0.98	89.81 ± 0.50	85.21 ± 1.32	91.20 ± 0.82	69.96 ± 2.94	93.84 ± 0.75	97.17 ± 0.30	**97.77 ± 0.26**

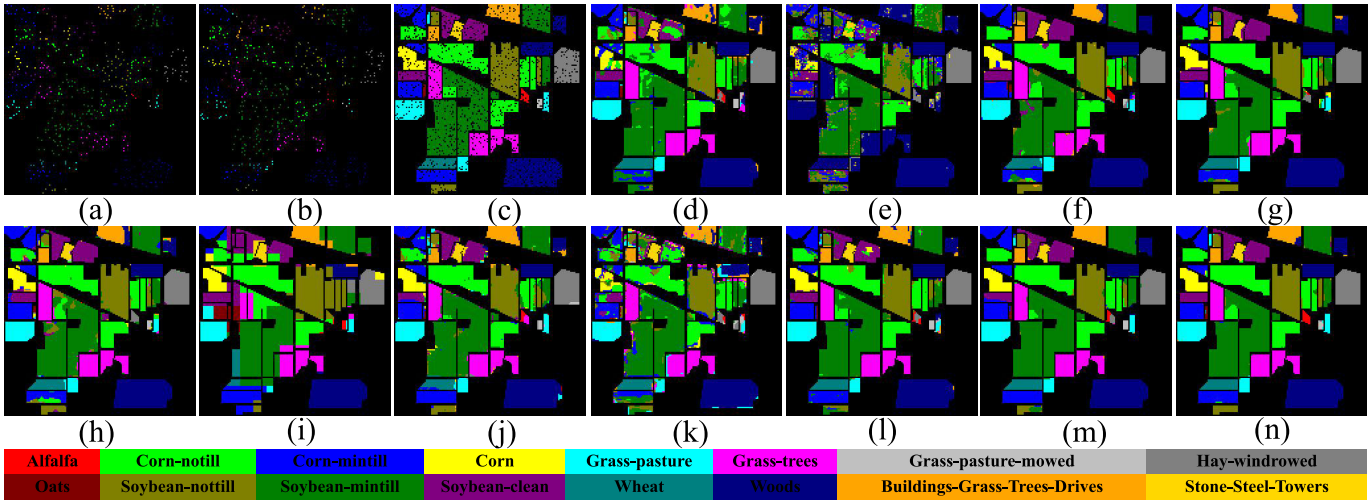


Fig. 8. Ground truth and classification maps obtained by different methods for Indian Pines dataset. (a) Training labels. (b) Validation labels. (c) Testing labels. (d) M3D-DCNN. (e) AB-LSTM. (f) A^2S^2KResNet. (g) SSRN. (h) HybridSN. (i) miniGCN. (j) VIT. (k) SpectralFormer. (l) GAHT. (m) SSFTT. (n) HyperSINet.

complexity primarily dominated by the LMOVE from the VIT branch. Within LMOVE, the quantity of self-attention and multihead computations determines the order of magnitude of computational complexity. Specifically, if the number of tokens is denoted as g, the time complexity of HyperSINet is $\mathcal{O}(g^2d + gd^2)$, where d represents the size of the hidden features.

C. Experimental Results

The classification accuracies of various models for each dataset are presented in Tables III–VI, while the classification maps achieved through these models are illustrated in Figs. 8–11.

1) Indian Pines Dataset: As evident from Table III, it is observed that AB-LSTM exhibits poor performance on classes 7 and 9, with notable variability in results across multiple runs, indicating instability in its performance. M3D-DCNN outperforms AB-LSTM overall. However, it demonstrates lower accuracy in classes with fewer samples, highlighting an imbalance in its classification performance. Both A^2S^2KResNet and SSRN outperform HybridSN in classification with the help of attention mechanisms. However, in Fig. 8, all five CNN-based models struggle with misclassifications in large areas, revealing their difficulty in handling nonlocal features in extensive spatial contexts. Mini-scale patches adversely affect miniGCN's performance, hindering its ability to handle large target regions. Among the four VIT-based methods, SpectralFormer exhibits suboptimal performance and struggles to effectively handle small-sample data. SSFTT, on the other hand, achieves higher accuracy compared with the other three methods due to its long-range modeling capabilities. However, HyperSINet, which takes a dual perspective by considering both nonlocal and local features, achieves the best classification performance and obtains a classification map closest to the ground truth.

2) PaviaU Dataset: As depicted in Table IV, while HybridSN achieves the highest classification accuracy in class 4, its OA is the lowest, indicating suboptimal performance in other categories. AB-LSTM and SpectralFormer exhibit comparable performance on this dataset, with AB-LSTM displaying lower stability compared with the latter. On the dataset, the performance of the five CNN-based models is subpar, further indicating the inherent limitations of using pure CNN models in capturing complex features. Based on the classification map displayed in Fig. 9, it is observed that CNN-based models tend to introduce salt-and-pepper noise

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TABLE IV

COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE PAVIAU DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	M3D-DCNN	AB-LSTM	A^{2}S^{2}KResNet	SSRN	HybridSN	miniGCN	VIT	SpectralFormer	GAHT	SSFTT	HyperSINet
class 1	86.97 ± 5.17	82.12 ± 2.95	81.50 ± 2.76	75.65 ± 5.11	72.79 ± 1.04	89.43 ± 0.80	86.39 ± 0.42	86.61 ± 4.92	95.31 ± 3.07	89.53 ± 1.16	**98.13 ± 0.89**
class 2	95.43 ± 2.65	94.39 ± 3.47	91.92 ± 0.88	93.28 ± 4.53	89.28 ± 1.24	98.55 ± 0.34	98.77 ± 0.29	92.06 ± 3.64	98.27 ± 1.26	**99.86 ± 0.13**	99.50 ± 0.26
class 3	49.72 ± 6.47	28.84 ± 8.42	81.43 ± 1.11	47.69 ± 0.49	59.03 ± 4.28	**90.59 ± 8.89**	79.21 ± 2.00	36.61 ± 20.10	74.64 ± 4.58	71.46 ± 3.67	88.94 ± 0.43
class 4	90.32 ± 5.78	87.50 ± 4.51	96.95 ± 3.12	97.46 ± 2.82	**98.30 ± 2.62**	64.70 ± 3.47	87.35 ± 1.26	77.80 ± 6.91	96.47 ± 1.91	87.83 ± 5.25	96.69 ± 0.71
class 5	98.86 ± 0.96	93.38 ± 11.23	98.76 ± 0.92	98.58 ± 0.38	96.21 ± 3.01	99.96 ± 0.06	99.97 ± 0.04	99.95 ± 0.09	**99.98 ± 0.03**	99.42 ± 0.46	99.75 ± 0.43
class 6	46.92 ± 9.81	39.89 ± 25.10	93.57 ± 0.05	89.71 ± 4.14	67.57 ± 0.82	**99.88 ± 0.09**	91.43 ± 1.04	53.01 ± 8.37	95.22 ± 1.78	93.36 ± 2.81	97.22 ± 1.47
class 7	66.68 ± 26.60	66.82 ± 12.91	91.18 ± 0.33	67.22 ± 5.48	48.05 ± 9.46	93.84 ± 0.67	97.54 ± 1.63	40.32 ± 14.94	81.56 ± 13.47	**98.37 ± 0.90**	81.31 ± 3.78
class 8	87.96 ± 3.78	83.95 ± 4.78	74.06 ± 0.18	72.69 ± 3.22	42.03 ± 10.20	92.49 ± 0.08	68.21 ± 3.37	70.24 ± 11.57	91.90 ± 8.33	85.48 ± 1.45	**94.04 ± 3.54**
class 9	97.58 ± 1.67	98.93 ± 0.69	84.21 ± 0.44	60.03 ± 10.35	46.97 ± 6.51	33.73 ± 5.44	84.85 ± 3.91	95.69 ± 0.39	99.25 ± 0.57	97.49 ± 0.25	**99.96 ± 0.06**
OA(%)	84.42 ± 0.85	80.68 ± 2.83	87.21 ± 0.04	84.98 ± 1.79	76.31 ± 1.35	92.42 ± 0.59	91.27 ± 0.27	79.72 ± 1.32	95.17 ± 0.40	93.89 ± 0.62	**97.28 ± 0.24**
AA(%)	80.05 ± 2.42	75.09 ± 3.28	88.17 ± 0.17	78.03 ± 0.33	68.91 ± 0.91	84.80 ± 1.35	88.19 ± 0.82	72.48 ± 2.34	92.51 ± 1.81	91.42 ± 0.89	**95.06 ± 0.46**
Kappa × 100	78.98 ± 1.08	73.84 ± 4.07	82.83 ± 0.06	79.77 ± 0.25	67.99 ± 0.71	89.95 ± 0.76	88.39 ± 0.35	72.66 ± 1.72	93.60 ± 9.54	91.85 ± 0.06	**96.40 ± 0.32**

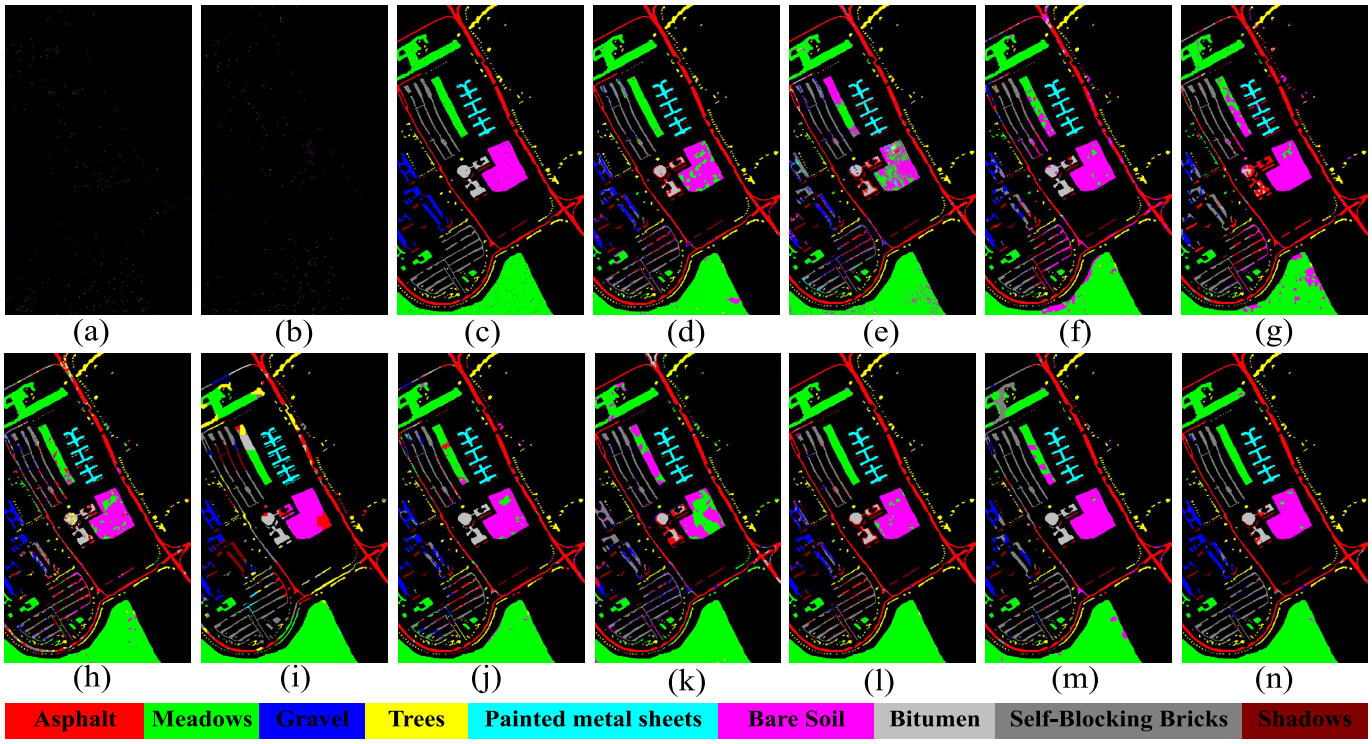


Fig. 9. Ground truth and classification maps obtained by different methods for the PaviaU dataset. (a) Training labels. (b) Validation labels. (c) Testing labels. (d) M3D-DCNN. (e) AB-LSTM. (f) A^{2}S^{2}KResNet. (g) SSRN. (h) HybridSN. (i) miniGCN. (j) VIT. (k) SpectralFormer. (l) GAHT. (m) SSFTT. (n) HyperSINet.

during the classification process. MiniGCN outperforms VIT and achieves optimal performance in three categories, but it also exhibits issues of class imbalance. GAHT, by introducing grouped pixel embeddings and benefiting from its global-local mechanism, surpasses SSFTT. With the assistance of a pair of interactors, HyperSINet surpasses GAHT by 2.11%, as indicated by the highest AA value, which suggests that HyperSINet has a well-balanced classification ability, making it highly suitable for both local and nonlocal features.

3) Salinas Dataset: As shown in Table V, HyperSINet obtained an accuracy of 97.88%, surpassing SSFTT by 1.53%. This further demonstrates the advantages of employing the dual-branch structure with interactors for feature processing. SSFTT outperforms VIT slightly but does not fully utilize its higher model complexity, potentially due to the single-branch structure's limitations in preserving fundamental features during processing. In the Grapes_untrained region, SSFTT exhibits higher misclassification rates and slightly lower classification stability, as indicated by the standard deviation, while the classification map in Fig. 10 also shows that miniGCN's excessive smoothing of features leads to blurred boundaries for specific land cover types, attributed to the effects of superpixel segmentation. Five CNN-based methods have demonstrated relatively stable classification performance, with A^{2}S^{2}KResNet exhibiting the best performance. A^{2}S^{2}KResNet, SSRN, and HybridSN achieve the highest classification accuracy in several categories. However, AB-LSTM performs the poorest in class 11. The classification maps presented in Fig. 10 indicate deficiencies in the classification capabilities of CNN-based methods, particularly in large target areas, revealing shortcomings in their global modeling capabilities.

4) Houston Dataset: As shown in Table VI, AB-LSTM continues to manifest notable instability on the dataset, contributing to significant variations in its classification performance across different categories. In contrast, M3D-DCNN demonstrates heightened classification capabilities due to the joint learning of 2-D multiscale spatial features and 1-D

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TABLE V

COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE SALINAS DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	M3D-DCNN	AB-LSTM	A^2S^2KResNet	SSRN	HybridSN	miniGCN	VIT	SpectralFormer	GAHT	SSFTT	HyperSINet
class 1	91.86 ± 6.94	53.34 ± 10.48	99.32 ± 0.59	**99.96 ± 0.17**	99.59 ± 0.05	91.24 ± 9.98	99.88 ± 0.20	92.64 ± 1.75	96.30 ± 4.21	99.79 ± 0.42	99.76 ± 0.49
class 2	98.67 ± 2.15	98.45 ± 0.18	99.60 ± 0.25	99.59 ± 0.06	95.48 ± 0.33	99.95 ± 0.06	**100.00 ± 0.00**	97.72 ± 2.30	99.64 ± 0.73	99.65 ± 0.42	99.69 ± 0.38
class 3	81.08 ± 7.27	43.66 ± 29.06	**99.92 ± 0.62**	89.60 ± 3.26	99.64 ± 0.05	99.17 ± 1.18	99.83 ± 0.08	85.57 ± 4.91	94.38 ± 3.68	99.83 ± 0.19	99.40 ± 0.71
class 4	94.10 ± 4.39	97.32 ± 0.18	96.75 ± 0.46	87.11 ± 2.89	95.66 ± 0.78	95.75 ± 1.68	95.77 ± 1.16	92.43 ± 4.70	97.29 ± 3.33	93.38 ± 10.35	**98.55 ± 1.31**
class 5	96.10 ± 1.78	96.15 ± 2.04	97.51 ± 0.60	98.22 ± 0.23	**99.84 ± 0.20**	87.96 ± 6.61	97.45 ± 2.16	82.14 ± 5.90	96.57 ± 3.60	98.79 ± 1.07	98.69 ± 1.19
class 6	99.64 ± 0.95	99.03 ± 0.52	99.76 ± 0.02	99.39 ± 0.06	99.67 ± 0.11	96.62 ± 2.55	99.86 ± 0.25	99.26 ± 0.71	**100.00 ± 0.00**	99.36 ± 0.90	99.87 ± 0.16
class 7	99.24 ± 0.48	99.29 ± 0.74	**100.00 ± 0.00**	99.77 ± 0.08	**100.00 ± 0.00**	98.26 ± 1.41	99.96 ± 0.07	97.56 ± 1.72	99.54 ± 0.70	99.47 ± 0.54	99.57 ± 0.50
class 8	83.25 ± 3.26	92.38 ± 11.15	88.00 ± 2.03	86.47 ± 3.22	93.08 ± 1.38	91.51 ± 2.82	93.50 ± 1.20	78.40 ± 5.60	89.69 ± 4.77	93.53 ± 10.89	**97.00 ± 1.88**
class 9	95.95 ± 1.80	98.63 ± 2.26	98.90 ± 0.36	99.00 ± 0.59	99.49 ± 0.35	**99.98 ± 0.04**	99.96 ± 0.04	94.28 ± 0.62	99.14 ± 0.68	**99.98 ± 0.01**	**99.98 ± 0.02**
class 10	83.37 ± 8.95	68.29 ± 17.29	97.93 ± 0.21	**99.35 ± 0.10**	98.60 ± 1.62	96.55 ± 2.60	98.94 ± 0.55	77.55 ± 7.43	94.51 ± 2.25	**99.35 ± 0.24**	97.73 ± 1.94
class 11	73.15 ± 8.14	0.00 ± 0.00	98.66 ± 0.06	96.63 ± 0.42	95.44 ± 1.96	93.12 ± 0.61	94.23 ± 4.34	72.22 ± 4.61	92.00 ± 6.38	**98.73 ± 1.33**	98.29 ± 2.39
class 12	98.48 ± 1.57	76.83 ± 11.43	98.21 ± 0.32	99.19 ± 0.71	98.22 ± 1.71	95.07 ± 4.22	**99.84 ± 0.05**	91.09 ± 5.83	99.60 ± 0.49	99.75 ± 0.48	99.74 ± 0.36
class 13	98.99 ± 1.43	31.20 ± 10.15	99.73 ± 0.09	91.30 ± 0.54	46.70 ± 16.23	60.43 ± 12.45	52.00 ± 17.23	87.67 ± 7.84	**99.85 ± 0.15**	56.85 ± 33.87	90.62 ± 18.82
class 14	94.72 ± 3.49	30.69 ± 20.58	88.65 ± 0.55	87.75 ± 0.76	25.76 ± 22.63	86.11 ± 8.59	94.11 ± 1.01	95.64 ± 3.00	**97.19 ± 2.02**	92.71 ± 2.56	97.03 ± 2.33
class 15	66.81 ± 7.36	3.99 ± 32.17	90.49 ± 0.23	86.44 ± 1.23	91.25 ± 0.94	**93.92 ± 5.52**	86.31 ± 1.98	73.74 ± 7.06	80.57 ± 7.74	93.22 ± 4.54	93.52 ± 2.78
class 16	75.06 ± 7.82	41.25 ± 17.70	99.94 ± 0.07	99.24 ± 0.03	**100.00 ± 0.00**	98.45 ± 1.84	98.14 ± 1.49	81.97 ± 9.55	90.46 ± 7.20	97.28 ± 3.62	99.04 ± 0.88
OA(%)	87.54 ± 0.98	71.88 ± 0.56	95.19 ± 0.03	92.59 ± 0.11	94.61 ± 0.21	94.47 ± 1.03	95.37 ± 0.33	85.95 ± 0.71	93.63 ± 0.29	96.35 ± 2.00	**97.88 ± 0.51**
AA(%)	89.40 ± 1.02	64.41 ± 0.32	97.09 ± 0.05	94.94 ± 0.05	89.90 ± 1.23	92.75 ± 1.72	94.36 ± 1.28	87.49 ± 0.82	95.42 ± 0.85	95.10 ± 1.61	**98.03 ± 0.93**
Kappa × 100	86.11 ± 1.09	68.34 ± 1.87	94.64 ± 0.02	91.74 ± 0.07	94.00 ± 0.38	93.85 ± 1.13	94.84 ± 0.37	84.37 ± 0.77	92.90 ± 0.33	95.94 ± 2.21	**97.02 ± 1.44**

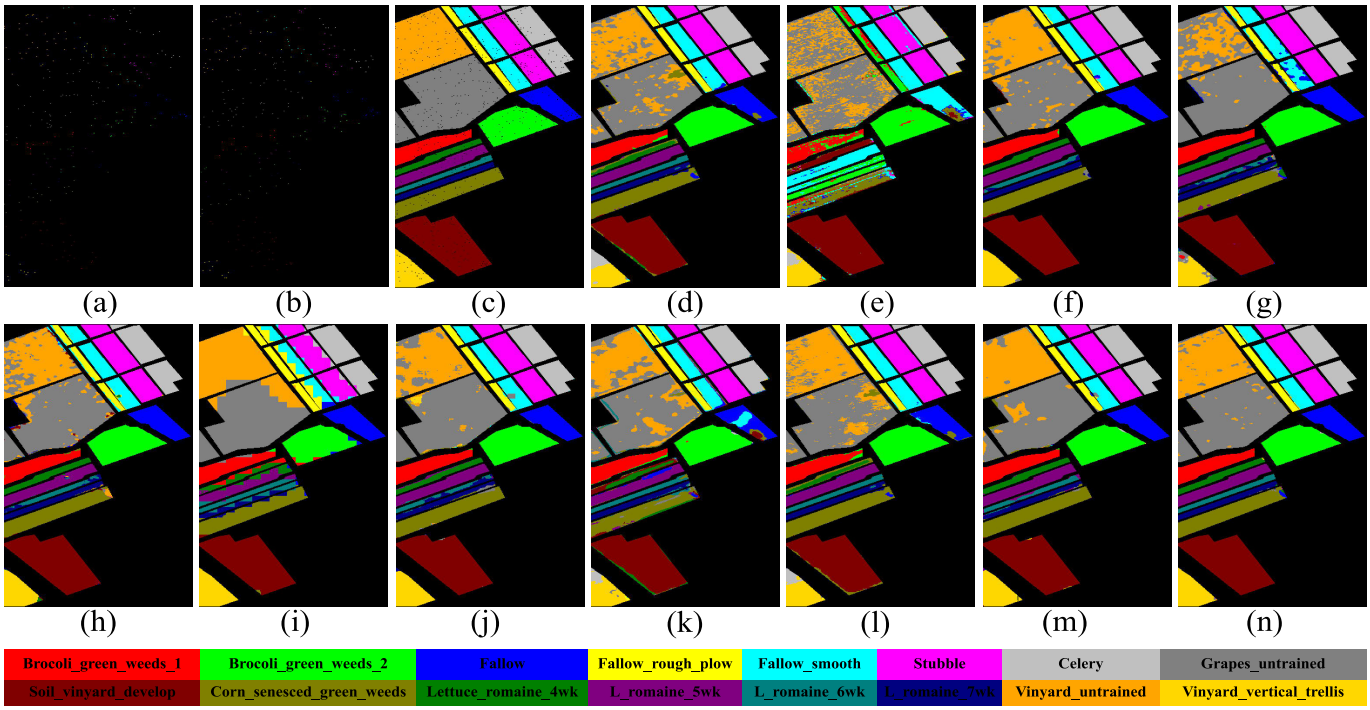


Fig. 10. Ground truth and classification maps obtained by different methods for the Salinas dataset. (a) Training labels. (b) Validation labels. (c) Testing labels. (d) M3D-DCNN. (e) AB-LSTM. (f) A^2S^2KResNet. (g) SSRN. (h) HybridSN. (i) miniGCN. (j) VIT. (k) SpectralFormer. (l) GAHT. (m) SSFTT. (n) HyperSINet.

spectral features. Among the other three CNN-based models, HybridSN shows the poorest performance, although it achieves 100% classification accuracy in class 14. The performance of A^2S^2KResNet and SSRN is comparable, indicating that attention mechanisms can to some extent compensate for the limitations of CNN in handling nonlocal features. Although serving as foundational models, miniGCN and VIT outperform SpectralFormer, achieving 100% accuracy in multiple categories. However, it is worth noting that due to mini-patch segmentation, miniGCN exhibits limited stability, leading to significant accuracy fluctuations. SSFTT shows high accuracy across five categories but is limited by its single-branch architecture. GAHT emphasizes local spectrum relationships, neglecting spatial connections, and limiting accuracy improvement. In contrast, HyperSINet's dual-branch design and interactors allow balanced use of local and nonlocal features, resulting in superior performance.

IV. DISCUSSION

In this section, we conduct various ablation experiments to demonstrate the roles of different modules in HyperSINet and the impact of certain hyperparameter settings. Additionally, we compare HyperSINet with several other DL models in terms of parameters, floating point operations per second (FLOPs), train time, and test time, aiming for a comprehensive evaluation of HyperSINet.

A. Determination of Segmentation Scale and Fusion Rate

The segmentation scale and dual-branch fusion rate are pivotal hyperparameters with a significant impact on OA. Given the relatively high time complexity of the VIT encoder, opting for a lower segmentation scale results in an excessive number of superpixels, particularly exerting a substantial computational demand, notably on the Houston

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TABLE VI

COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE HOUSTON DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	M3D-DCNN	AB-LSTM	A^2S^2KResNet	SSRN	HybridSN	miniGCN	VIT	SpectralFormer	GAHT	SSFTT	HyperSINet
class 1	96.93 ± 1.27	91.55 ± 5.22	98.02 ± 0.01	96.09 ± 0.03	89.91 ± 1.45	94.92 ± 4.43	97.81 ± 0.46	92.27 ± 3.66	98.19 ± 0.39	**99.18 ± 1.48**	98.42 ± 1.43
class 2	98.23 ± 1.48	91.09 ± 5.87	99.14 ± 0.13	98.08 ± 0.01	98.81 ± 0.12	99.10 ± 0.40	99.32 ± 0.41	96.71 ± 1.27	98.78 ± 1.34	99.57 ± 0.35	**99.59 ± 0.22**
class 3	99.55 ± 0.74	97.16 ± 0.78	**100.00 ± 0.00**	**100.00 ± 0.00**	97.17 ± 0.27	98.90 ± 1.51	**100.00 ± 0.00**	98.41 ± 0.76	**100.00 ± 0.00**	99.78 ± 0.21	**100.00 ± 0.00**
class 4	97.54 ± 2.54	91.84 ± 0.96	97.98 ± 0.01	98.05 ± 0.01	89.53 ± 1.64	94.19 ± 4.08	95.74 ± 2.98	94.30 ± 2.49	97.25 ± 2.34	98.21 ± 1.42	**99.30 ± 0.96**
class 5	99.45 ± 0.80	97.30 ± 0.53	99.84 ± 0.32	99.95 ± 0.11	95.56 ± 0.34	98.49 ± 2.54	99.88 ± 0.10	98.00 ± 1.17	**100.00 ± 0.00**	99.98 ± 0.04	99.77 ± 0.32
class 6	91.37 ± 7.14	81.78 ± 2.12	**100.00 ± 0.00**	97.97 ± 0.01	96.13 ± 0.32	98.71 ± 2.23	**100.00 ± 0.00**	75.75 ± 6.76	93.29 ± 5.20	**100.00 ± 0.00**	98.63 ± 2.34
class 7	89.76 ± 3.49	82.26 ± 2.58	96.94 ± 0.12	97.68 ± 0.04	88.28 ± 1.46	93.14 ± 4.28	94.80 ± 0.57	82.30 ± 2.39	94.06 ± 1.63	96.00 ± 0.44	**97.77 ± 0.49**
class 8	85.36 ± 4.42	59.32 ± 4.90	99.48 ± 0.01	**99.59 ± 0.02**	83.60 ± 1.33	90.99 ± 6.57	96.19 ± 0.14	83.64 ± 3.10	93.18 ± 3.02	95.48 ± 1.35	96.52 ± 1.71
class 9	89.57 ± 1.07	78.81 ± 2.57	95.19 ± 0.02	95.40 ± 0.03	76.89 ± 5.33	88.01 ± 9.66	88.97 ± 3.33	77.91 ± 3.17	94.82 ± 2.00	95.31 ± 1.26	**97.20 ± 0.92**
class 10	91.63 ± 1.15	75.92 ± 9.71	93.02 ± 0.14	91.43 ± 0.02	90.63 ± 0.72	92.38 ± 6.21	94.93 ± 6.69	85.05 ± 3.18	96.97 ± 1.45	**99.86 ± 0.32**	99.42 ± 0.47
class 11	82.84 ± 5.43	69.44 ± 1.98	95.23 ± 0.04	96.75 ± 0.02	93.00 ± 0.95	92.03 ± 1.59	94.01 ± 1.76	72.05 ± 1.70	96.35 ± 1.61	97.52 ± 4.12	**99.66 ± 0.27**
class 12	92.61 ± 2.24	85.24 ± 4.38	97.17 ± 0.02	92.53 ± 0.02	94.58 ± 0.82	97.50 ± 2.64	97.96 ± 0.58	90.81 ± 3.69	96.74 ± 0.93	**99.69 ± 0.08**	99.39 ± 0.47
class 13	92.65 ± 1.85	37.87 ± 13.68	93.69 ± 0.03	95.30 ± 0.04	83.18 ± 1.97	84.28 ± 5.30	89.73 ± 0.55	50.62 ± 7.85	**95.73 ± 0.62**	90.57 ± 6.12	94.97 ± 1.27
class 14	96.16 ± 2.74	87.38 ± 9.44	99.02 ± 0.02	97.72 ± 0.03	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	83.27 ± 5.32	99.74 ± 0.23	**100.00 ± 0.00**	**100.00 ± 0.00**
class 15	99.70 ± 0.31	96.57 ± 2.15	99.09 ± 0.01	99.57 ± 0.02	92.26 ± 1.77	97.42 ± 4.47	**100.00 ± 0.00**	97.00 ± 1.43	**100.00 ± 0.00**	97.27 ± 1.99	98.58 ± 1.72
OA(%)	93.15 ± 0.67	82.36 ± 2.44	97.34 ± 0.03	96.77 ± 0.03	90.43 ± 0.03	94.32 ± 3.37	96.33 ± 0.19	86.74 ± 0.74	96.92 ± 0.33	97.98 ± 0.90	**98.67 ± 0.19**
AA(%)	93.56 ± 0.79	81.57 ± 2.66	97.59 ± 0.04	97.08 ± 0.12	91.30 ± 0.11	94.67 ± 2.92	96.62 ± 0.16	85.21 ± 0.99	97.01 ± 0.40	97.90 ± 0.94	**98.61 ± 0.20**
Kappa × 100	92.59 ± 0.72	80.92 ± 2.64	97.14 ± 0.01	96.50 ± 0.04	89.65 ± 0.06	93.86 ± 3.65	96.03 ± 0.20	85.66 ± 0.80	96.67 ± 0.36	97.81 ± 0.97	**98.57 ± 0.20**

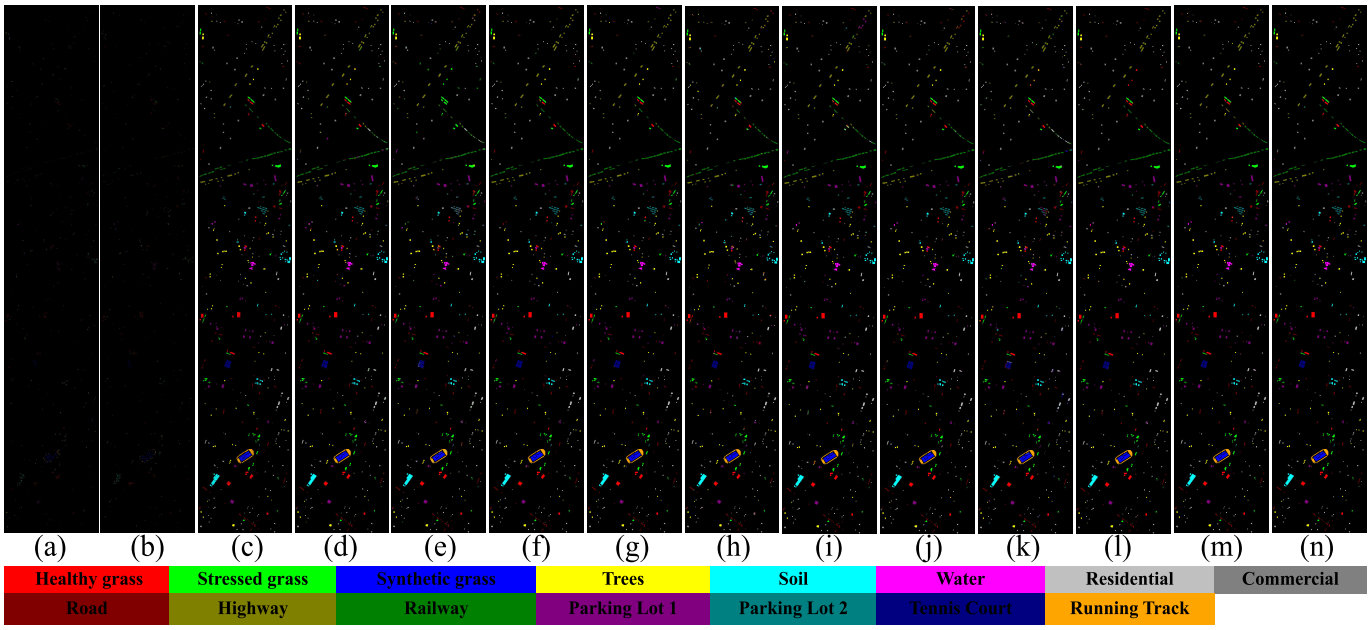


Fig. 11. Ground truth and classification maps obtained by different methods for Houston dataset. (a) Training labels. (b) Validation labels. (c) Testing labels. (d) M3D-DCNN. (e) AB-LSTM. (f) A^2S^2KResNet. (g) SSRN. (h) HybridSN. (i) miniGCN. (j) VIT. (k) SpectralFormer. (l) GAHT. (m) SSFTT. (n) HyperSINet.

dataset. Consequently, following meticulous experimentation, we have established the lower segmentation scale as 300, and the upper segmentation scale as 700, with increments of 100 between them. The set of fusion rate values comprises [0, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50]. There are a total of 35 combinations for each dataset, and the results are illustrated in Fig. 12, where the highest classification accuracy is denoted by green-colored dots.

In Fig. 12, it is noteworthy that for the Indian Pines, PaviaU, and Houston, the highest accuracy is achieved with a segmentation scale of 500 and fusion rate of 0.05, respectively. For the Salinas, the optimal combination is (600, 0.05). The choice of segmentation scale influences the number of pixels within each superpixel. In the case of Salinas, due to the conflicting nature of its spatial dimensions and label distribution, larger segmentation scales are more appropriate to avoid covering too much local information. Conversely, lower segmentation scales work well for the other three datasets. To enhance the robustness of HyperSINet, a segmentation scale of 500 is adopted. As for the fusion rate, experiments across all four datasets consistently favor a rate of 0.05, which is, therefore, chosen as the fusion rate.

B. Comparison of Different Training Sample Ratio

In our experiments, we set the training sample ratios to 0.05 for Indian Pines and Houston and 0.005 for PaviaU and Salinas. However, different training sample ratios can lead to varying model performances. To systematically evaluate HyperSINet, we employ different training sample ratios. Specifically, for Indian Pines and Houston, we use the training sample ratios of [0.05, 0.06, 0.07, 0.08, 0.09, 0.10], while for the other two datasets, the training sample ratios are set as [0.005, 0.006, 0.007, 0.008, 0.009, 0.010], respectively. The OA values of the 11 models on four datasets under various training sample ratios are displayed in Fig. 13.

As depicted in Fig. 13, AB-LSTM exhibits relatively poor performance across all four datasets. Despite an increase in OA with rising label rates, the improvement in precision is limited when compared with the other four CNN-based

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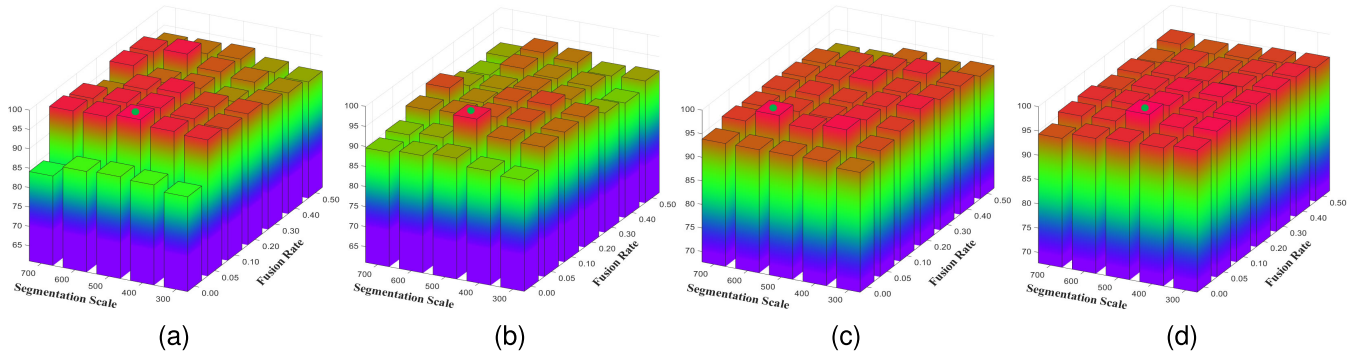


Fig. 12. Impact of different combinations of segmentation scale and fusion rate on the OA. (a) Indian Pines. (b) PaviaU. (c) Salinas. (d) Houston.

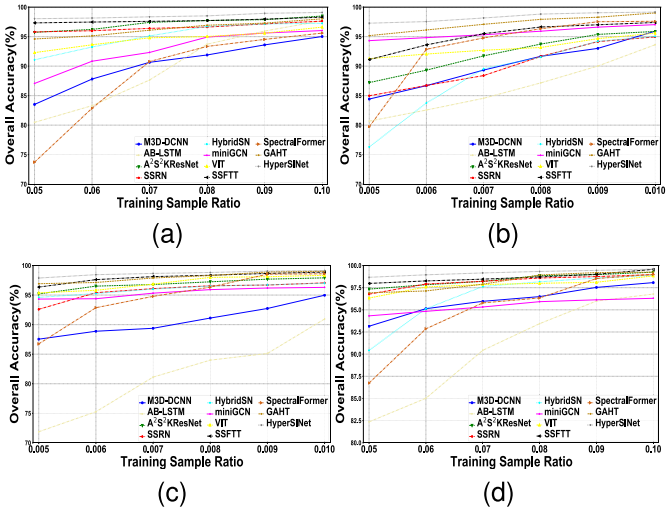


Fig. 13. Impact of Training Sample Ratios on the OA. (a) Indian Pines. (b) PaviaU. (c) Salinas. (d) Houston.

models. On the other hand, while M3D-DCNN outperforms AB-LSTM, its performance is constrained by inherent model limitations, indicating an insufficient upper bound on accuracy. Among the other three CNN-based models, A^2S^2KResNet achieves the most prominent performance, mainly due to the integration of attention mechanisms and nonfixed receptive fields. Furthermore, across three datasets, the CNN-based models consistently outperform miniGCN as the ratio increases, indicating that CNN-based models have a higher performance ceiling than GNN-based models with the increase in training sample ratios. Even though miniGCN shows a slight upward trend in OA with an increase in the ratio, the trend is not significantly prominent. The performance of VIT is inferior to that of SSFTT but superior to miniGCN, further demonstrating the limitations of GNN-based models. While SpectralFormer exhibits comparatively weaker performance at low label ratios, its performance improves significantly as the label ratio increases. However, it struggles to surpass the performance of SSFTT and GAHT. GAHT, which models both global and local spectral information and employs a grouping approach, enhances model performance to a level comparable to that of SSFTT. HyperSINet demonstrates the highest level of classification performance across various training sample ratios, validating its strong classification capability. By employing a dual-branch structure, HyperSINet effectively extracts both local and nonlocal features. The interactors facilitate the integration of these two types of features during the training process, leading to improved classification and enhanced model performance.

C. Ablation Analysis

1) Ablation Study of the Proposed Modules: HyperSINet comprises multiple modules and seven sets of ablation experiments were conducted to assess their impact on the performance of HyperSINet, using OA as the evaluation metric. The experimental results are presented in Table VII. It is noteworthy that the Baseline refers to HyperSINet without the inclusion of the two branches.

Table VII reveals that when employing only the Baseline, a mere linear classifier proves inadequate in capturing the intricate features present in HSIs, resulting in subpar performance across all four datasets. The second and third sets of experiments, respectively, validated the enhancements brought by the VIT branch and the CNN branch to the baseline. It can be observed that with the incorporation of either branch, HyperSINet's performance undergoes a significant enhancement, and the improvement capabilities of the two branches are comparable. The fourth set of experiments represents the performance of HyperSINet when none of the interactors is employed, surpassing the performance of both the second and third sets on all four datasets. Upon the separate incorporation of Conv2Trans and Trans2Conv, a further performance improvement is observed compared with the standalone dual-branch structure, substantiating the viability of embedding bridges within the dual-branch structure. The final set of experiments represents the empirical performance of the proposed HyperSINet. The performance on all four datasets surpasses that of the remaining six configurations, thereby substantiating the effectiveness of all components.

2) Ablation Study of the Fusion Strategy: To achieve more effective fusion in both Conv2Trans and Trans2Conv, a sigmoid-based fusion strategy has been introduced into both interactors, alleviating feature incompatibility issues associated with elementwise addition or elementwise multiplication. Fig. 14 illustrates the diagrammatic representations of three fusion strategies. Independent experiments are conducted for each strategy on four different datasets, with the OA serving

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TABLE VII
ABLATION ANALYSIS OF DIFFERENT MODULES ACROSS FOUR DATASETS

Baseline	Local Mask-Optimized VIT Encoder	CNN block	Trans2Conv	Conv2Trans	OA			
					Indian Pines	PaviaU	Salinas	Houston
✓	✗	✗	✗	✗	67.94 ± 2.98	72.32 ± 1.71	74.09 ± 3.75	63.17 ± 2.80
✓	✓	✗	✗	✗	92.59 ± 0.08	93.29 ± 0.58	93.06 ± 0.81	94.16 ± 0.97
✓	✗	✓	✗	✗	92.93 ± 0.12	92.67 ± 0.21	94.19 ± 0.67	94.06 ± 0.38
✓	✓	✓	✗	✗	95.09 ± 1.02	94.86 ± 1.23	95.19 ± 1.31	94.76 ± 1.58
✓	✓	✓	✗	✓	97.03 ± 1.76	96.78 ± 1.01	95.92 ± 1.39	95.68 ± 1.44
✓	✓	✓	✓	✗	97.19 ± 2.00	96.64 ± 1.32	96.18 ± 1.15	95.42 ± 0.77
✓	✓	✓	✓	✓	**98.04 ± 0.22**	**97.28 ± 0.24**	**97.88 ± 0.51**	**98.67 ± 0.19**

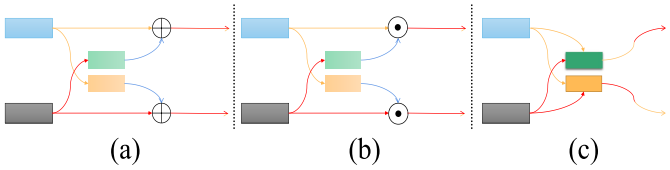


Fig. 14. Diagrammatic representation of three fusion strategies. (a) Elementwise addition strategy. (b) Elementwise multiplication strategy. (c) Proposed strategy.

TABLE VIII
ABLATION ANALYSIS OF DIFFERENT FUSION STRATEGIES ACROSS FOUR DATASETS

	Indian Pines	PaviaU	Salinas	Houston
Addition	95.92 ± 0.51	94.12 ± 0.23	93.98 ± 0.47	96.86 ± 0.34
Multiplication	88.03 ± 0.83	92.59 ± 0.67	95.78 ± 1.07	87.43 ± 1.40
Proposed	**98.04 ± 0.22**	**97.28 ± 0.24**	**97.88 ± 0.51**	**98.67 ± 0.19**

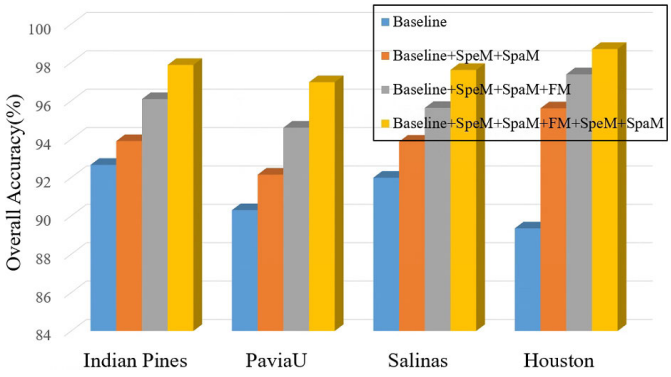


Fig. 15. Results of Ablation Experiments in the CNN block.

TABLE IX
ABLATION ANALYSIS OF DIFFERENT MASKS ACROSS FOUR DATASETS

	Indian Pines	PaviaU	Salinas	Houston
SNN mask	95.34 ± 1.26	93.07 ± 2.09	93.46 ± 1.90	95.21 ± 1.87
unnormalized mask	96.70 ± 0.67	95.27 ± 1.02	96.08 ± 0.95	96.43 ± 1.28
local mask	**98.04 ± 0.22**	**97.28 ± 0.24**	**97.88 ± 0.51**	**98.67 ± 0.19**

as the evaluation metric. It is noted that the interactors in Fig. 14(a) and (b) consist exclusively of local and nonlocal feature transformation stages. The results are presented in Table VIII.

As depicted in Table VIII, the addition strategy outperforms the multiplication strategy across three datasets, except Salinas. The addition strategy, to a certain extent, enables the integration of two features, whereas the multiplication strategy exhibits more pronounced feature incompatibility when fusing features. The two strategies do not take into account the distribution of the features themselves, thereby overlooking internal relationships. In contrast, the proposed sigmoid-based fusion strategy employs soft-selective weighted fusion, determining fusion weights based on the features themselves. The approach facilitates a more comprehensive feature fusion, resulting in higher performance.

3) Ablation Study of the Modules in CNN Block: In the CNN block, several key modules, namely, TM, SpeM, SpaM, and FM, exist. Four sets of ablation experiments are designed and conducted on each dataset to investigate the effectiveness of each module, with the results presented in Fig. 15.

The baseline in Fig. 15 represents the CNN block consisting only of two TMs, and the classification results are shown to be the worst on all datasets. However, with the introduction of one SpeM and one SpaM, the OA exhibits a notable improvement, indicating the effectiveness of extracting spectral and spatial features for the classification. The FM facilitates a more effective fusion of spectral and spatial features, enabling a more comprehensive exploration of data distributions, thus further improving the OA. The second set of SpeM and SpaM conducts a more in-depth exploration of the features output by FM, unearthing latent characteristics better suited for classification, and the OA reaches the highest.

4) Ablation Study of the Local Mask: In the VIT branch, we design an LMOVE embedded with a local mask, where the local mask is a normalized weighted adjacency matrix. To verify the effect of the local mask, we embed the SNN mask in LESSFormer, the unnormalized local mask (unnormalized mask), and the local mask into HyperSINet, and the results are shown in Table IX.

As demonstrated in Table IX, the OA of the local mask employed in HyperSINet ranks highest, followed by the unnormalized mask, with the SNN mask exhibiting the lowest performance. The SNN mask adopts a binary form, resulting in uniform weights assigned to all neighboring nodes and thereby diminishing discriminative capability. After normalization, the weights are constrained within the range of 0 to 1, mitigating adverse effects introduced by singular values and consequently further enhancing OA.

To intuitively observe the effect of the local mask, attention maps of LMOVE are visualized, as shown in Fig. 16. The results presented in Fig. 16(a) depict the outcome without the incorporation of a local mask. It can be observed that

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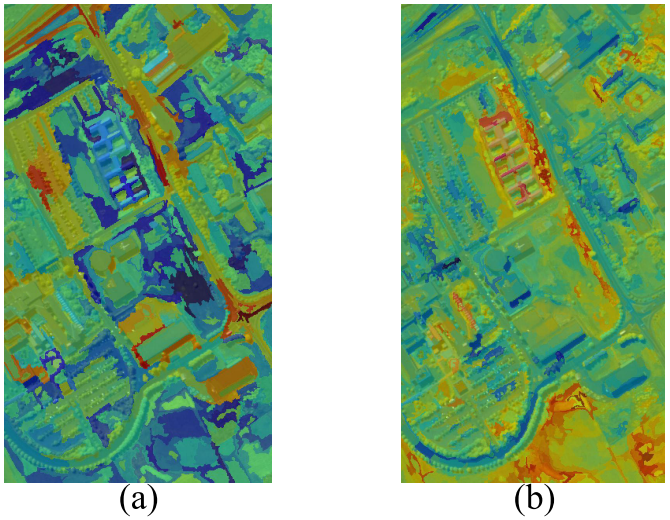


Fig. 16. Visualization of attention maps on the PaviaU. (a) Without local mask. (b) With local mask.

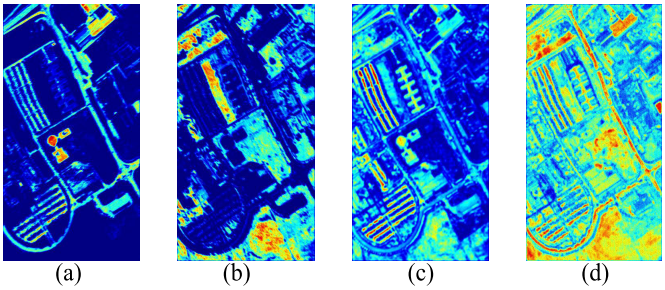


Fig. 17. Typical feature maps on the PaviaU. (a) Without both. (b) Only with Trans2Conv. (c) Only with Conv2Trans. (d) With all.

the refinement of local features is insufficient, and attention does not entirely cover all regions. Under the influence of the local mask, as illustrated in Fig. 16(b), attention extends to all details, emphasizing nonlocal aspects while simultaneously considering local features.

D. Visualization for the Function of Two Interactors

Conv2Trans and Trans2Conv interactors serve as bridges between two branches, facilitating the mutual transformation of nonlocal and local features. To visually illustrate the roles of the two interactors, we visualize the typical feature maps in four scenarios, i.e., without using any interactor, using only Trans2Conv, using only Conv2Trans, and using both interactors. The four typical feature maps are depicted in Fig. 17.

In Fig. 17, the intensity of colors corresponds to the level of activation in the image regions. Deeper colors indicate higher levels of activation. Fig. 17(a) is a typical feature map when no interactor is used. It can be seen that the activated regions are notably small, primarily concentrated in the Bitumen area and specific regions of self-blocking bricks, reflecting the limited feature extraction capability of the dual-branch network when no interactor is introduced. Upon the sole introduction of Trans2Conv, as depicted in Fig. 17(b), the highly activated regions expand, preserving the integrity of objects within large target areas. However, in smaller target

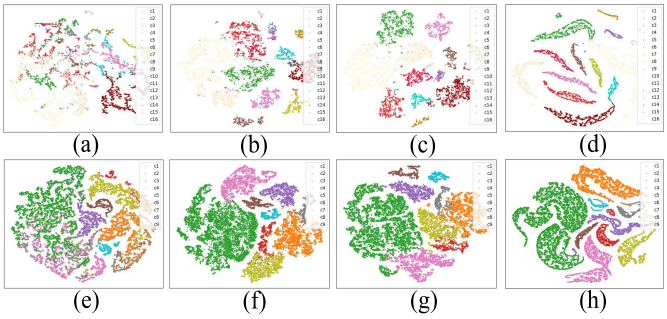


Fig. 18. T-SNE visualization of VIT-based models. (a) and (e) SpectralFormer for Indian Pines and PaviaU. (b) and (f) SSFTT for Indian Pines and PaviaU. (c) and (g) GAHT for Indian Pines and PaviaU. (d) and (h) HyperSINet for Indian Pines and PaviaU.

regions, such as the local details of gravel, the portrayal is not as pronounced. In contrast, Fig. 17(c) demonstrates the sole introduction of Conv2Trans, where PaviaU's local details are depicted with great precision, which proves highly effective in distinguishing between the painted metal sheets and asphalt regions. However, in large target areas, the integrity of objects is comparatively diminished. Fig. 17(d) presents the typical feature map resulting from the simultaneous incorporation of two interactors. It can be observed that, with the introduction of the two interactors, there is a qualitative improvement in both the depiction of local details and the integrity of nonlocal objects. A substantial portion of the image exhibits heightened activation.

E. Comparative T-SNE Visualization of VIT-Based Models

To provide a comprehensive evaluation of HyperSINet, T-Stochastic neighbor embedding (T-SNE) [58] visualization is conducted on its high-dimensional output features concerning Indian Pines and PaviaU. The results are presented in Fig. 18.

In the context of the Indian Pines, it is observed that SpectralFormer encounters challenges in its classification capabilities, leading to notable issues with excessive interclass similarity and reduced intraclass similarity. The challenges are particularly prominent in the classes c2, c6, and c12. SSFTT presents significant improvements in interclass discrimination. However, some degree of interclass overlap remains, especially in the classes c9 and c14. Additionally, SSFTT's intraclass cohesion is slightly lower, indicating considerable feature differences within the same class. GAHT shows significant discriminative performance across different classes with fewer interclass overlaps. However, it still exhibits relatively lower intraclass cohesion, leading to a visually scattered distribution of data points. Fig. 18(d) illustrates that HyperSINet achieves maximized interclass distinctiveness and intraclass cohesion, providing visual confirmation of its classification performance.

On the PaviaU, SpectralFormer demonstrates improved discrimination across different classes, with some residual confusion observed in c3 and c7. Moreover, data points within c4 and c5 exhibit substantial dispersion, indicating inadequate intraclass cohesion. GAHT, in comparison to SSFTT, achieves superior interclass distinctiveness and higher interclass cohesion. However, there is still insufficient intraclass cohesion within c8, and instances of confusion between c1 and c3.

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TABLE X
COMPARISON OF COMPUTATIONAL COST AND RUNNING TIME OF COMPETITORS ACROSS FOUR DATASETS

Dataset	Metrics	M3D-DCNN	AB-LSTM	A^2S^2K-ResNet	SSRN	HybridSN	miniGCN	VIT	Spectral Former	GAHT	SSFTT	HyperSINet
Indian Pines	P(M)	0.27	1.58	0.37	0.36	3.55	0.14	0.21	0.34	1.04	0.93	0.13
	F(MMac)	66.66	41.57	171.22	158.00	59.47	307.21	8.50	35.41	50.83	9.53	7.41
	Train time(s)	60.42	67.82	73.28	90.26	54.63	10.03	10.35	130.78	16.29	13.75	38.46
	Test time(s)	1.37	1.88	1.74	0.98	1.25	0.12	13.57	200.11	36.40	35.78	0.04
PaviaU	P(M)	0.08	0.52	0.22	0.22	1.76	0.12	0.20	0.16	0.93	0.48	0.12
	F(MMac)	31.97	17.81	87.70	81.02	29.04	210.32	6.43	14.44	45.41	4.91	14.98
	Train time(s)	24.72	28.80	15.09	15.11	16.15	24.29	12.97	90.64	13.17	11.94	53.28
	Test time(s)	11.09	14.29	7.17	5.29	4.94	0.23	25.68	180.57	49.06	35.68	0.10
Salinas	P(M)	0.29	1.64	0.37	0.37	3.62	0.14	0.21	0.35	0.97	0.95	0.13
	F(MMac)	67.71	42.70	174.70	161.21	60.73	110.93	8.50	36.43	47.61	9.72	11.15
	Train time(s)	39.64	48.98	49.12	98.93	30.01	16.51	15.44	150.34	15.16	13.20	46.78
	Test time(s)	7.09	10.01	9.17	6.26	5.74	0.17	27.78	341.89	39.19	37.01	0.11
Houston	P(M)	0.18	0.88	0.28	0.28	2.52	0.16	0.21	0.23	0.97	0.67	0.12
	F(MMac)	46.74	27.03	124.24	114.70	41.90	347.78	8.21	22.42	47.67	6.86	89.97
	Train time(s)	75.69	93.37	239.02	148.41	28.93	39.26	16.29	180.63	18.62	15.83	288.63
	Test time(s)	39.66	52.38	10.51	8.11	2.02	0.78	30.31	1980.56	70.89	60.19	0.20

In contrast, the visualization of HyperSINet reveals limited confusion only within c4 and c7, accompanied by significant enhancements in both interclass distinctiveness and intraclass cohesion.

F. Comparison of Computational Cost and Running Time

It is not comprehensive to solely evaluate the performance of a model based on metrics, such as OA, AA, Kappa, and per-class classification accuracy, as it does not involve an analysis of the model's feasibility. Therefore, we compare HyperSINet with ten other DL methods used in this article in terms of parameters, FLOPs, train time, and test time, aiming to provide a comprehensive evaluation. The comparative results are displayed in Table X.

As observed from Table X, AB-LSTM exhibits a higher parameter count compared with M3D-DCNN. However, it registers fewer FLOPs, attributed to M3D-DCNN's increased computational load due to the incorporation of numerous convolutional operations. Furthermore, AB-LSTM requires more training and testing time than M3D-DCNN, a mismatch with its lower classification performance. Besides, it can be observed that A^2S^2KResNet and SSRN have a similar number of parameters. However, A^2S^2KResNet exhibits slightly higher FLOPs. The disparity can be attributed to the more complex network architecture of A^2S^2KResNet, which consequently results in longer train and test times compared with SSRN. HybridSN exhibits the highest parameter count due to the utilization of two 3-D convolution layers. However, its network architecture is relatively simple, resulting in significantly lower values for FLOPs, train time, and test time compared with the other two CNN-based models. It is worth noting that the three models employed an early stopping strategy during the training process, whereby the train time, to some extent, reflects the speed of model convergence. However, it is possible that insufficient training could result from this strategy, hindering the attainment of the models' highest performance. Exhibited across the four datasets, miniGCN demonstrates the lowest parameter count and the highest FLOPs. The underlying rationale for the GNN-based methodology requires extensive computations involving matrices, thus leading to a significant number of FLOPs. In the VIT-based models, we have noted that except for HyperSINet, the testing time for all other models surpasses the training time. It is primarily attributed to the fact that the number of patches employed for training constitutes a minuscule fraction, while the majority of patches are allocated for testing. In the case of HyperSINet, the input data remain consistent between the training and testing phases, thereby resulting in the testing time being nearly equivalent to the time required for a single epoch of training.

V. CONCLUSION

Due to the lack of interaction between local and nonlocal features in the traditional dual-branch networks based on VIT and CNN, a potential issue of incompatibility may arise during the final fusion. Therefore, in this article, we propose a dual-branch synergetic interaction network, named HyperSINet, which is built upon VIT and CNN branches. HyperSINet achieves synergistic interaction between nonlocal and local features through a pair of interactors, thereby enabling the collaboration between the VIT and CNN branches. Additionally, within the VIT branch, an optimized VIT encoder incorporating a local mask is devised to refine the local features. In the CNN branch, a specialized CNN block is formulated to capture spectral and spatial local details, effectively enhancing the model's feature representation capacity. Experimental results on four real-world datasets demonstrate that HyperSINet exhibits superior classification performance compared with other state-of-the-art models.

In future work, we will further explore the design of more lightweight models for the domain. Specifically, we plan to employ a U-Net structure combined with VIT and CNN, integrating interactors in a manner that achieves efficient classification of HSIs in a low-parameter fashion.

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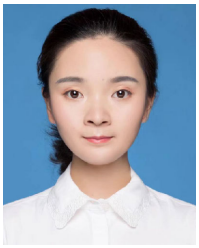
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